

Master Software Architecture

From Simplicity to Complexity

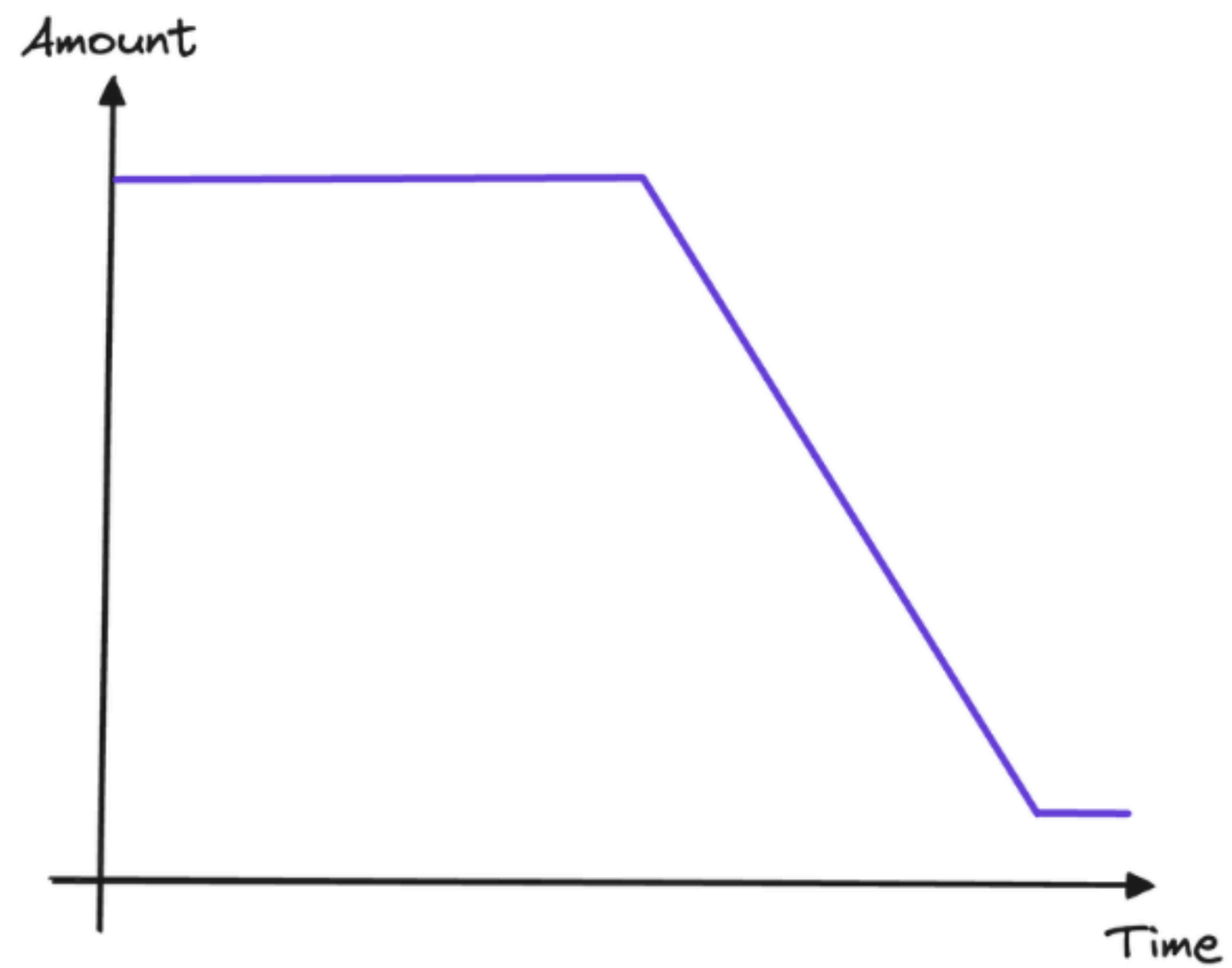
DOWNLOAD & USE **THE GOTO GUIDE APP**

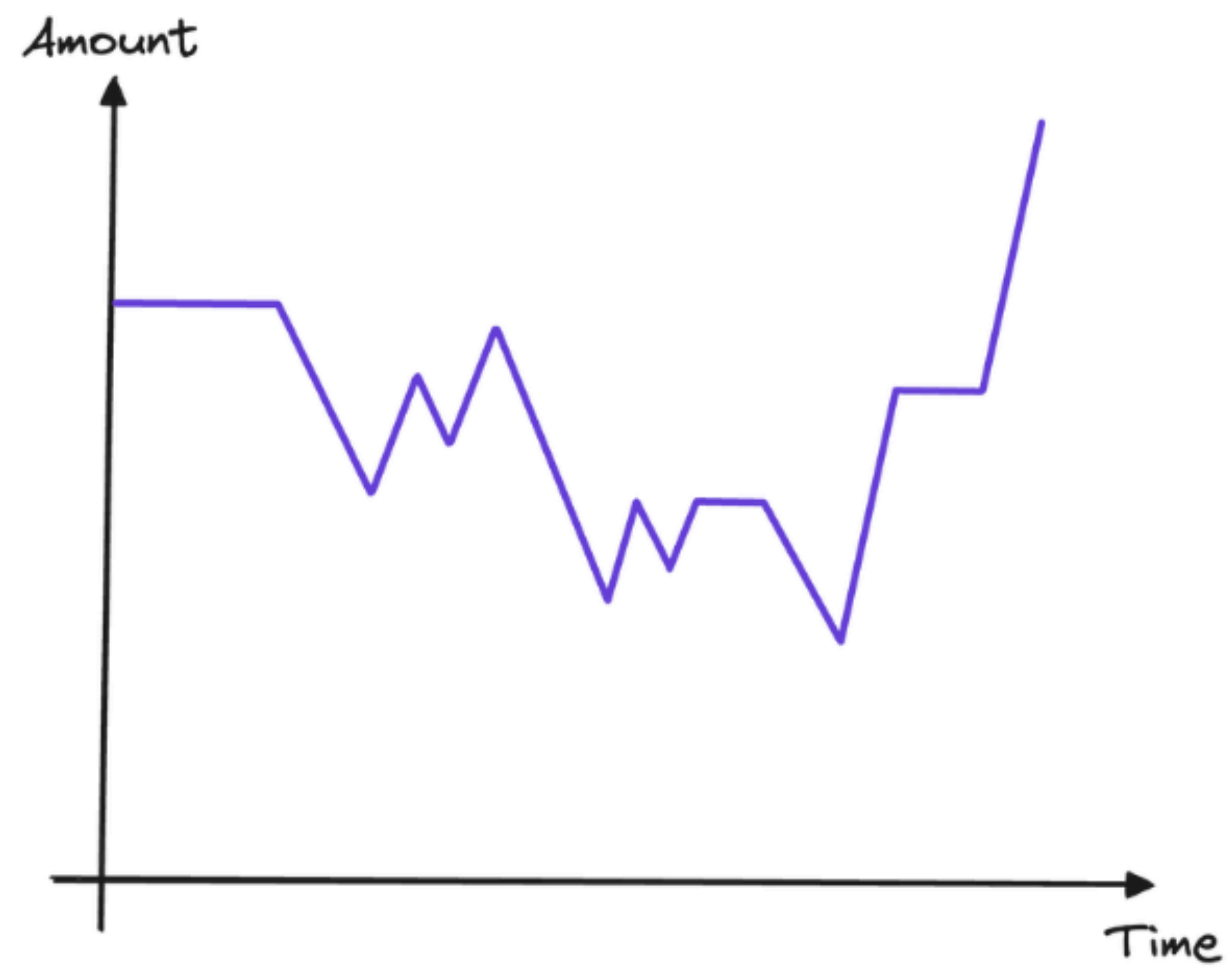
DOWNLOAD THE APP
CREATE YOUR SCHEDULE
ASK QUESTIONS
RATE SESSIONS



“Architecture is a set of design decisions
that are **hard to make** and even **harder to change.**”

-Vlad Khononov

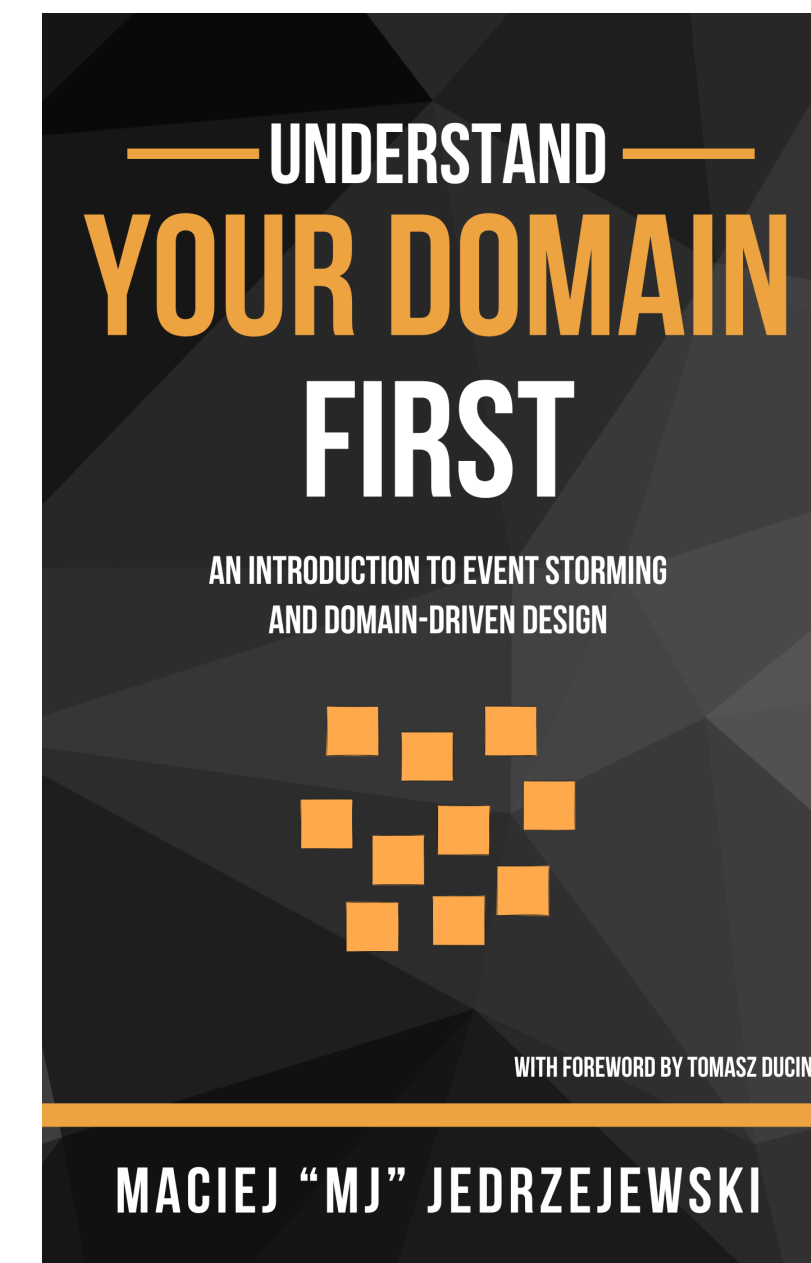
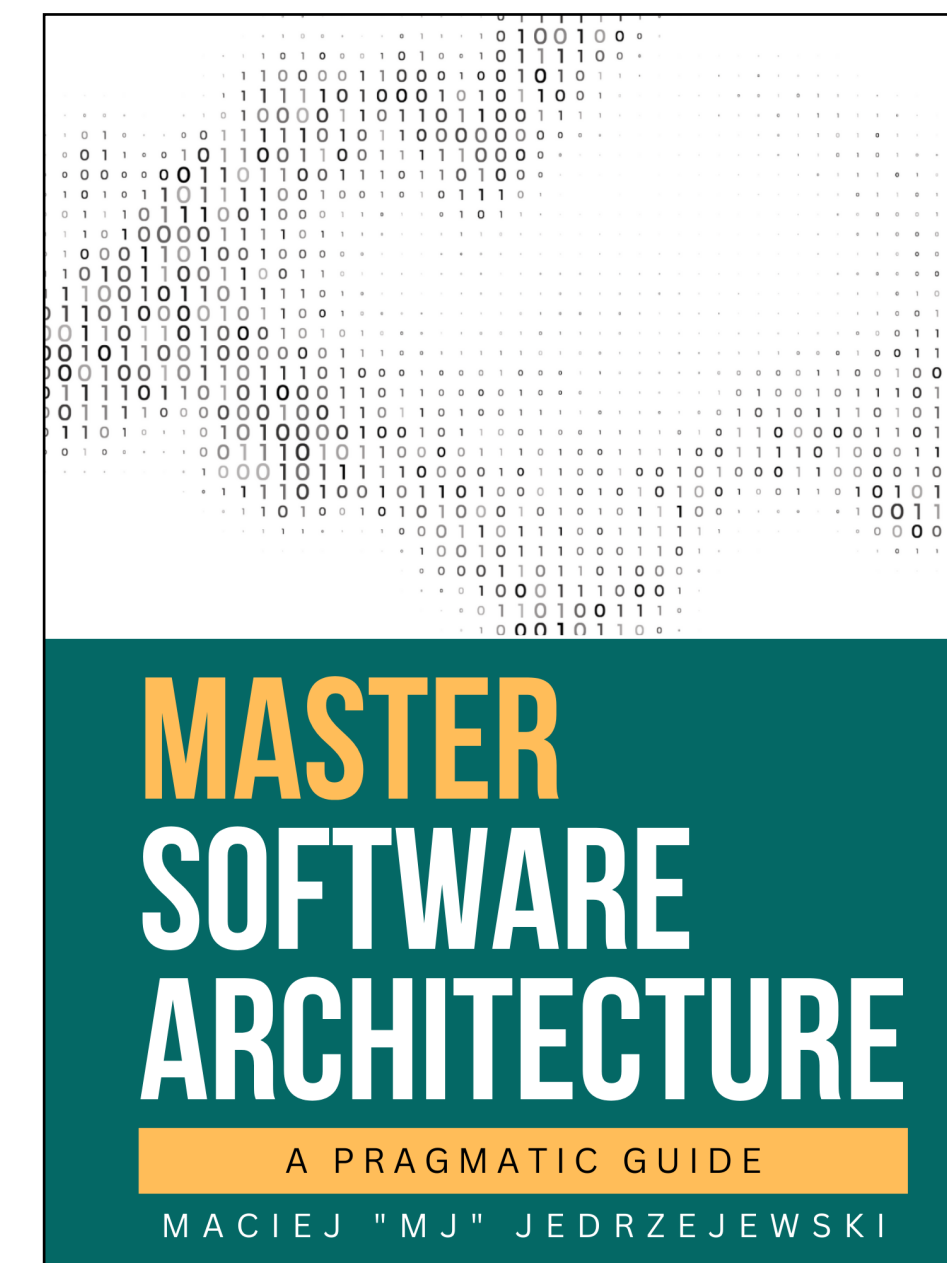




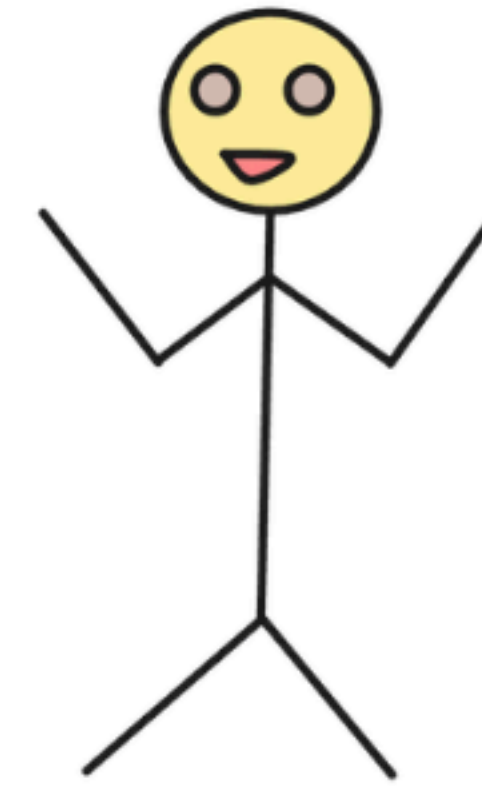
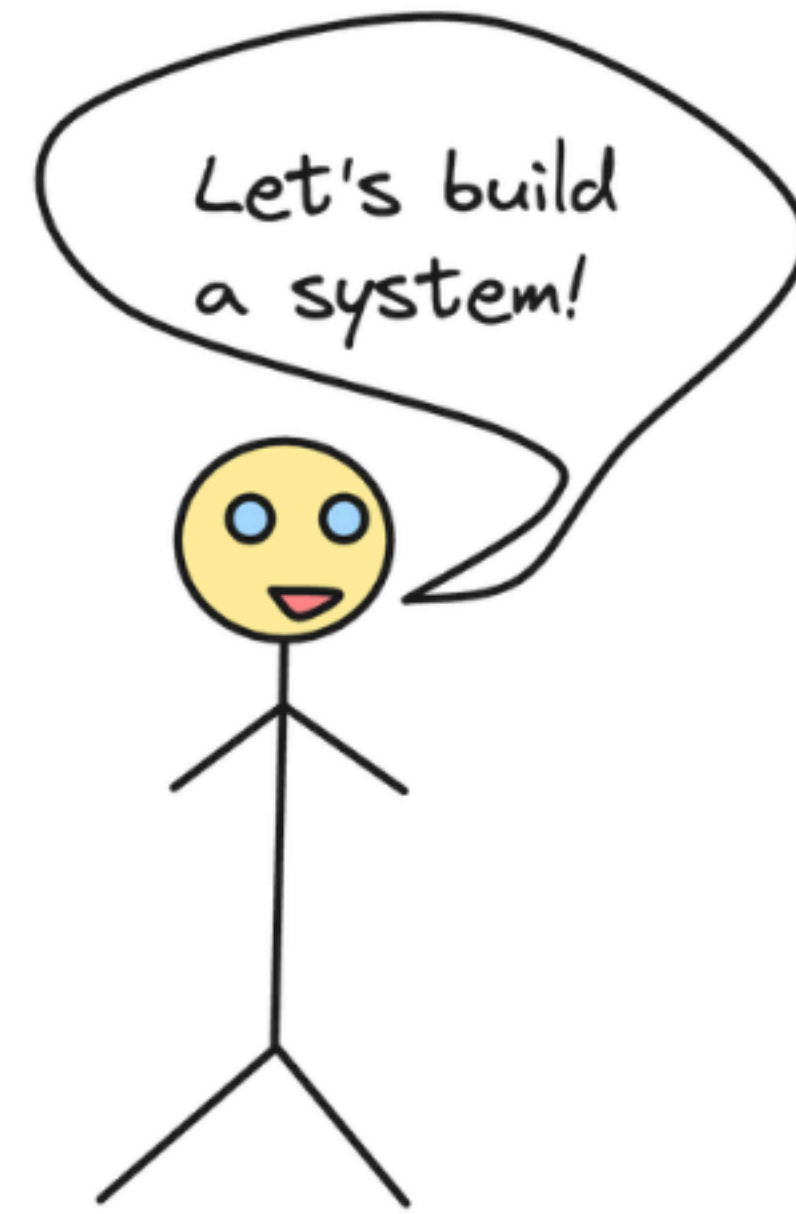


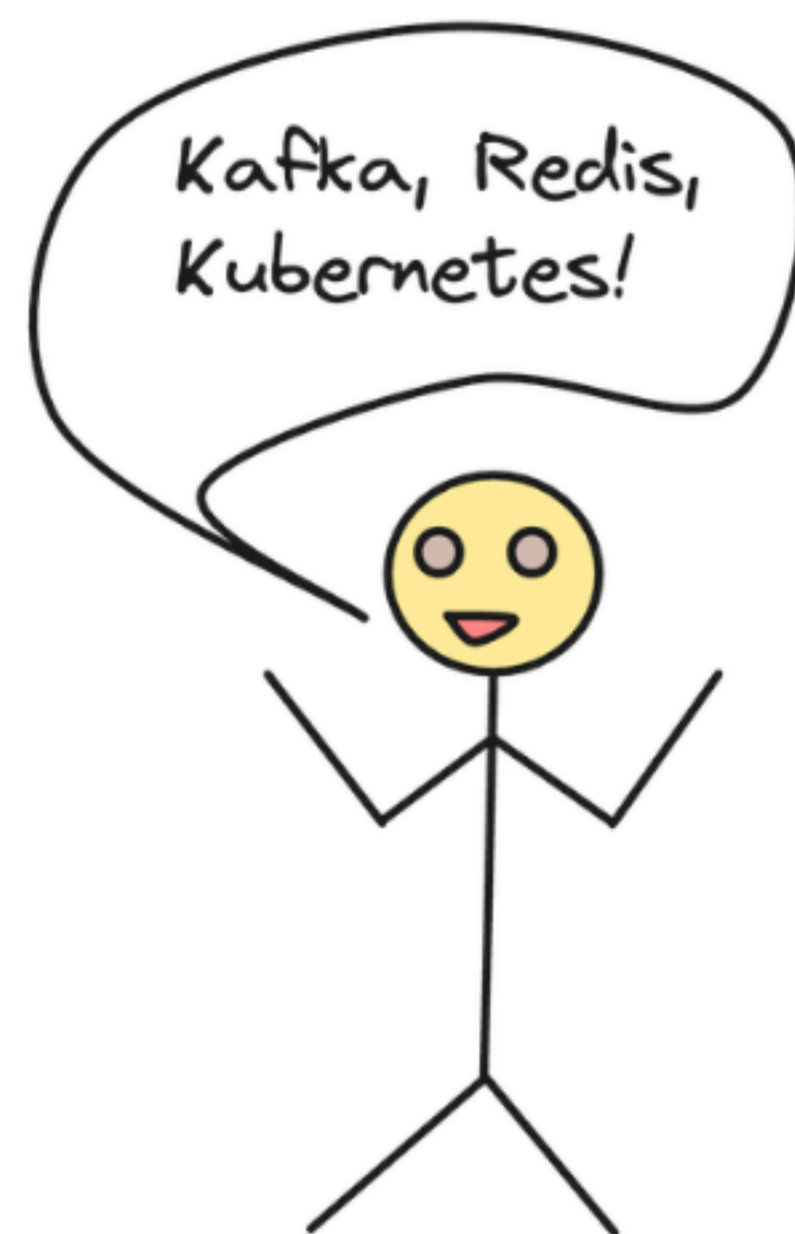
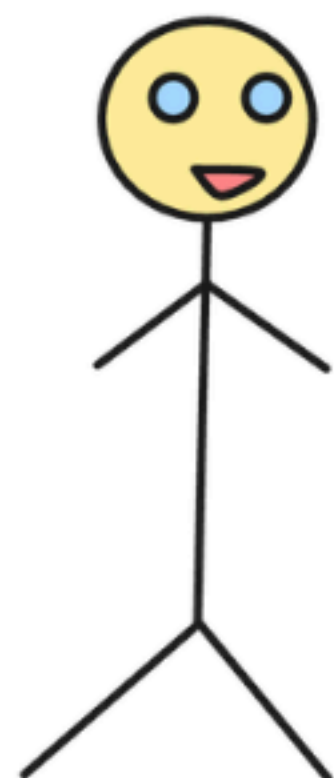
Team Lead | Fractional Architect

- 13 years in IT
- Trainer
- Domain-Driven Design evangelist
- Living in Switzerland



Somewhere in the Swiss Alps...





Why is this happening?

Cognitive shortcuts

FOMO

Over-engineered systems

Silver bullet

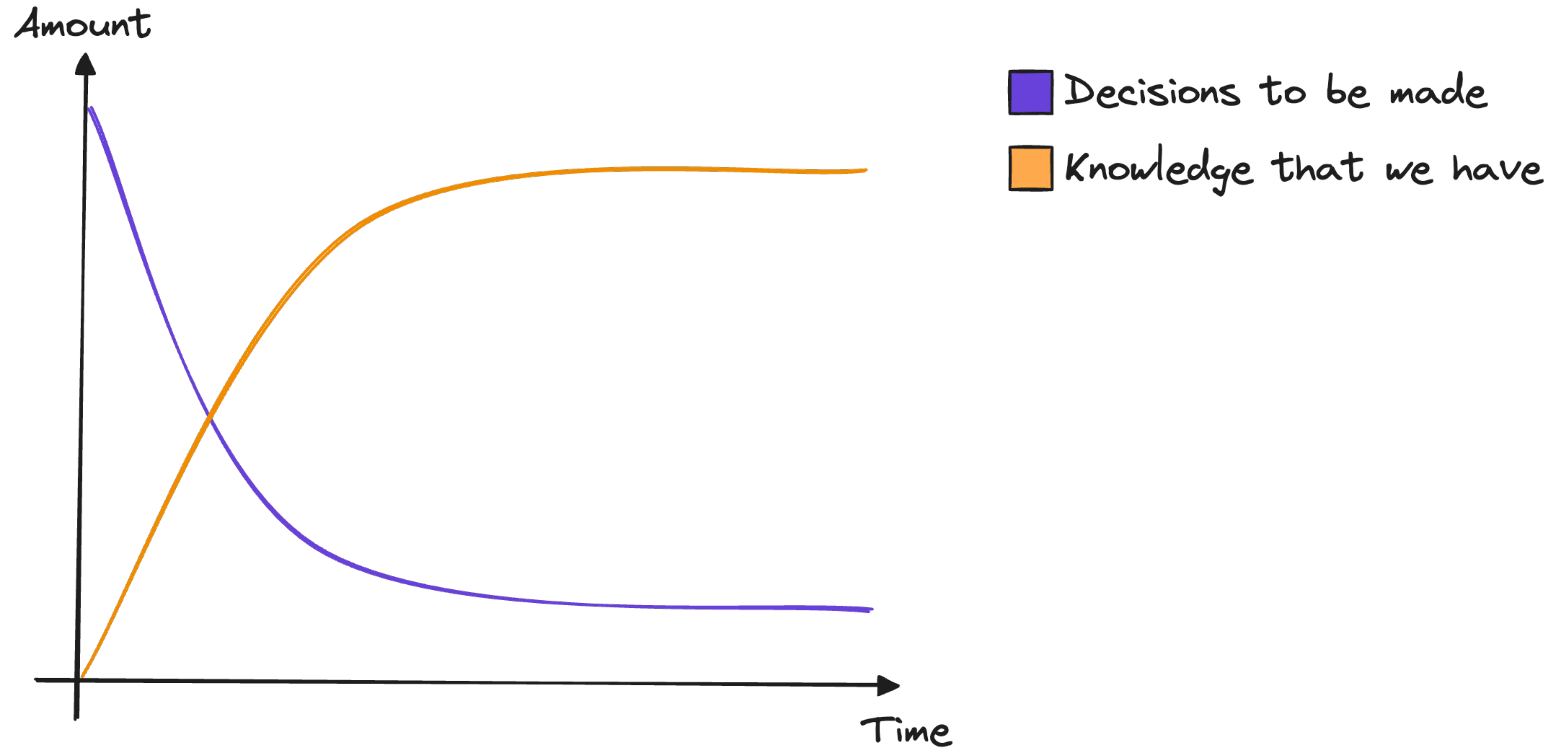
Swiss Army knife

Perfect design...

Golden key

...does not exist.

Context **is the king.**



Discovering your domain

First **listen**. Then **talk**.

- **Why** do we want to build this product?
- **What** problem are we trying to solve?
- **How** big is the market demand?

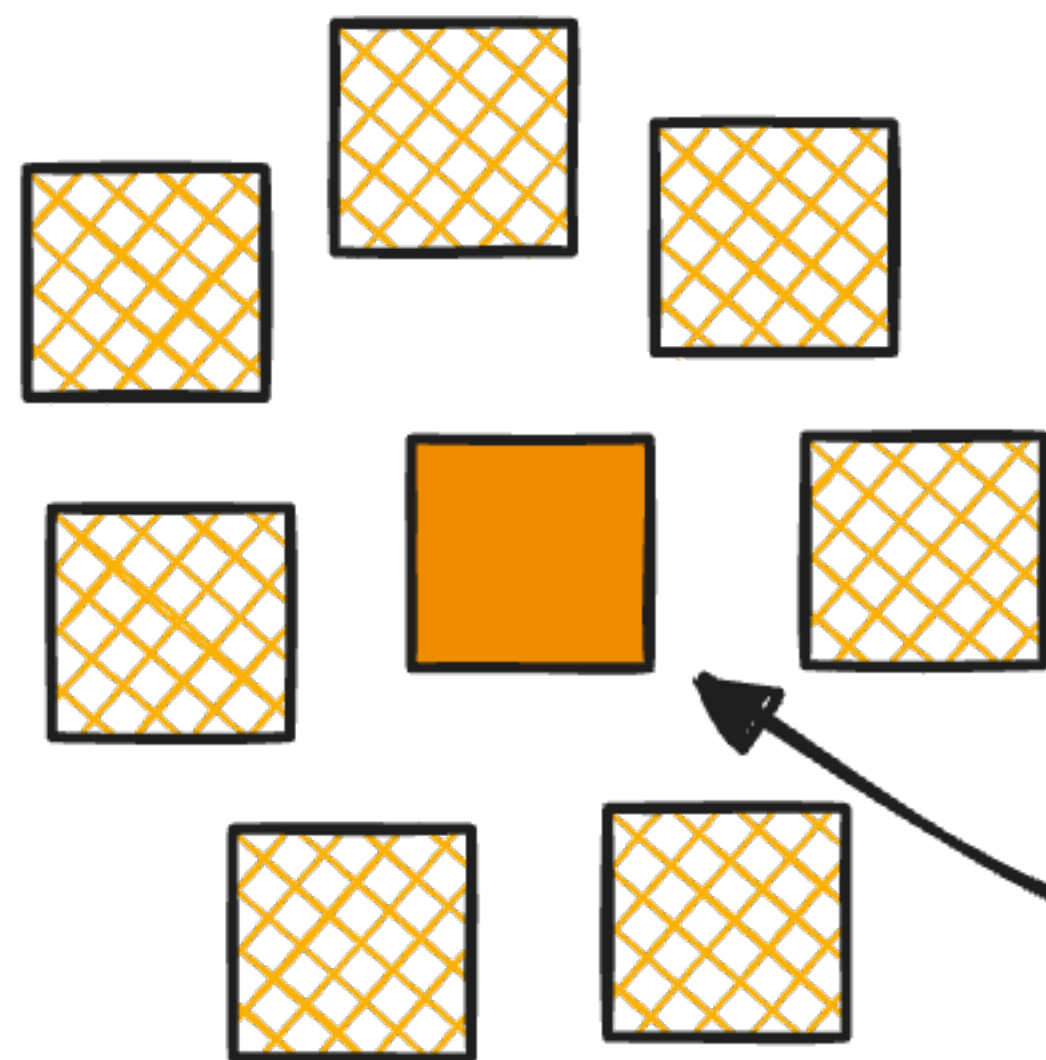
Oversimplification





How to minimize the risk?

Participants stick
events around
the central one



Place the central
event on the wall

Offer
prepared

Inquiry sent
for an offer

Offer
published

Offer
reviewed

Offer
approved

Pass
canceled

Contract
prepared

Offer
corrected

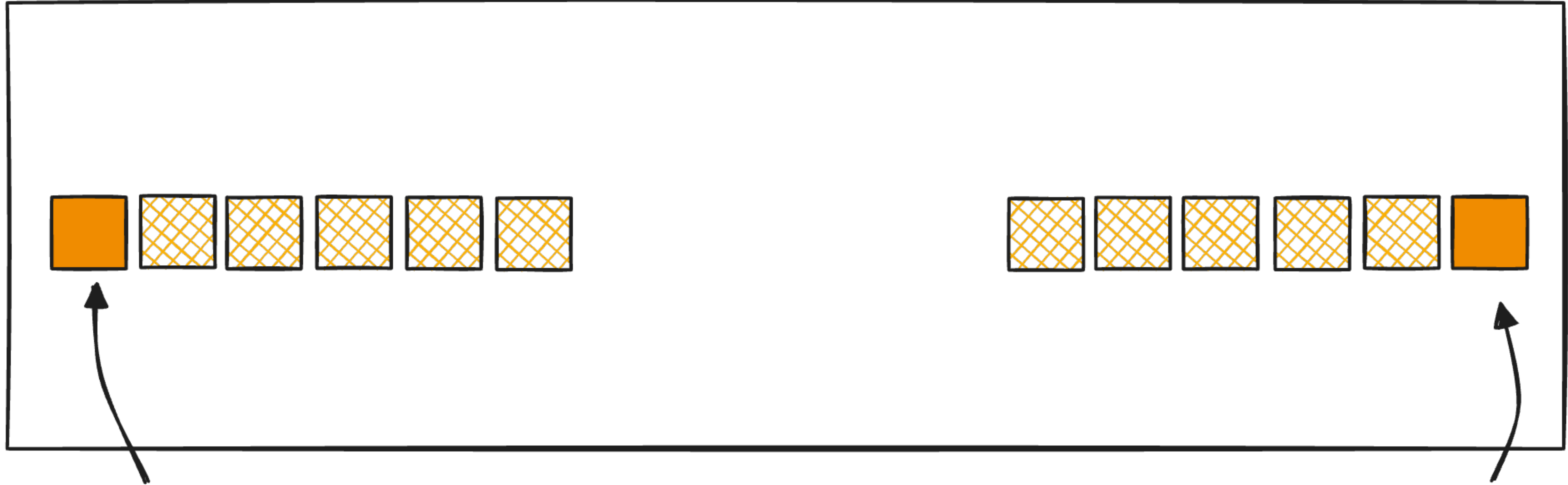
Pass
expired

Contract
expired

Contract
signed

Pass
registered

Contract
started



Start from the
beginning of the process
and move forward

Start from the
end of the process
and go backwards



Offering

It must be approved by a sales person

Inquiry sent for an offer

Inquiring

Contracting

Contract prepared

Contract signed

Payment gateway

Contract started

Contract expired

When contract is considered signed?

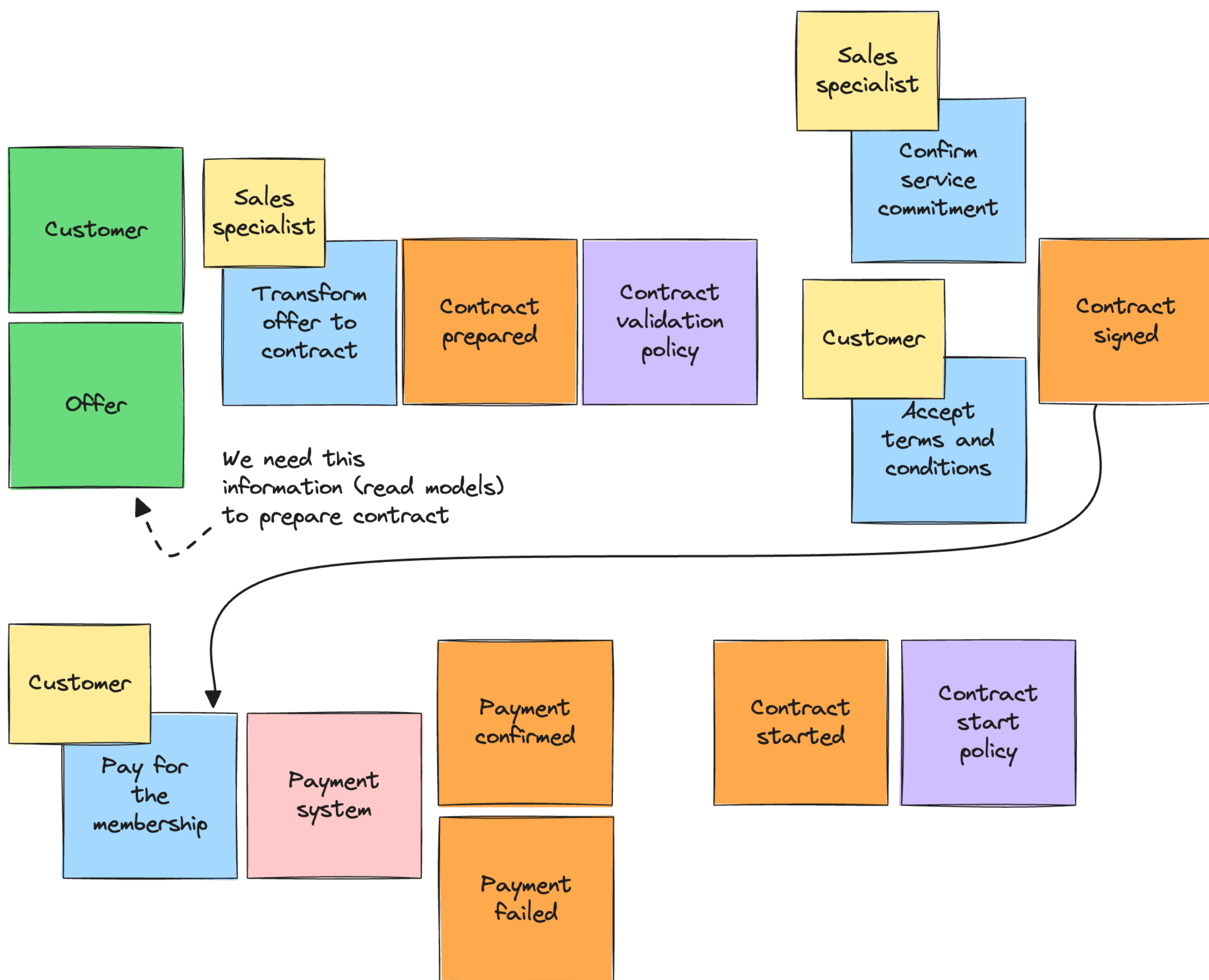
Pass Registration

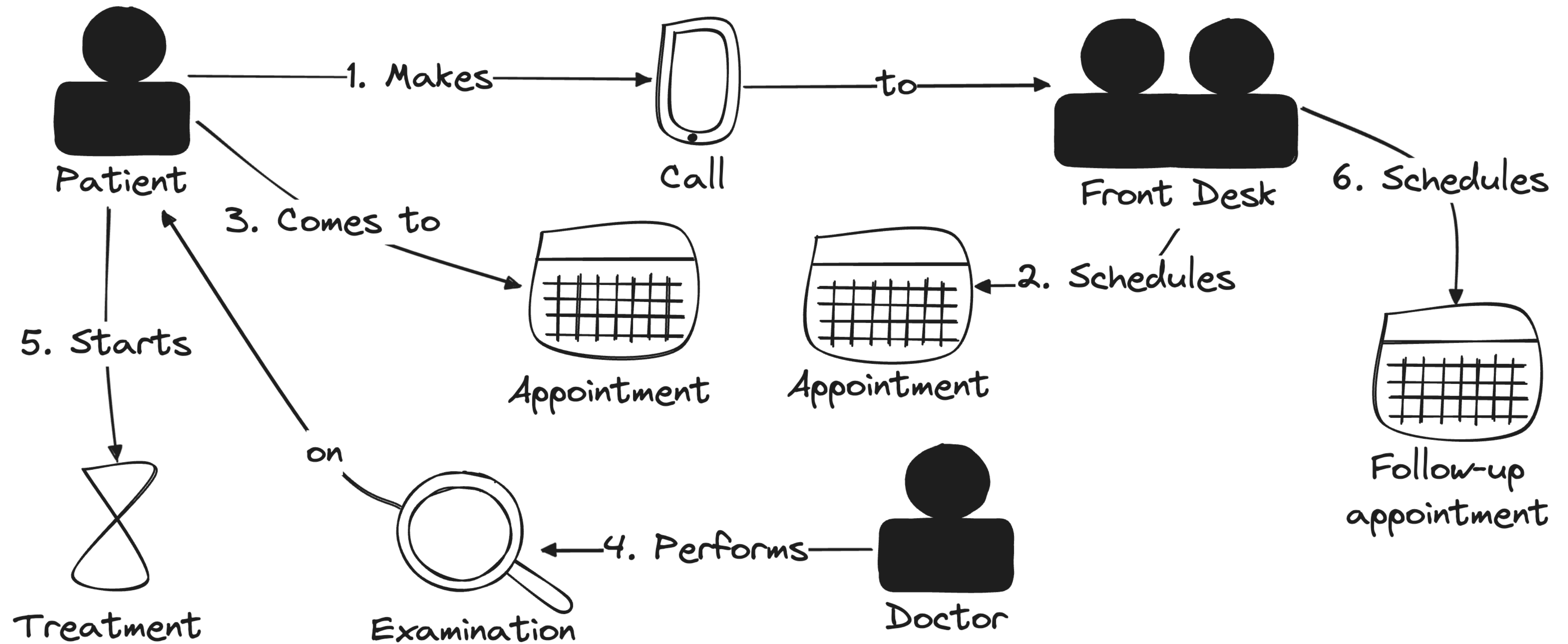
Pass registered

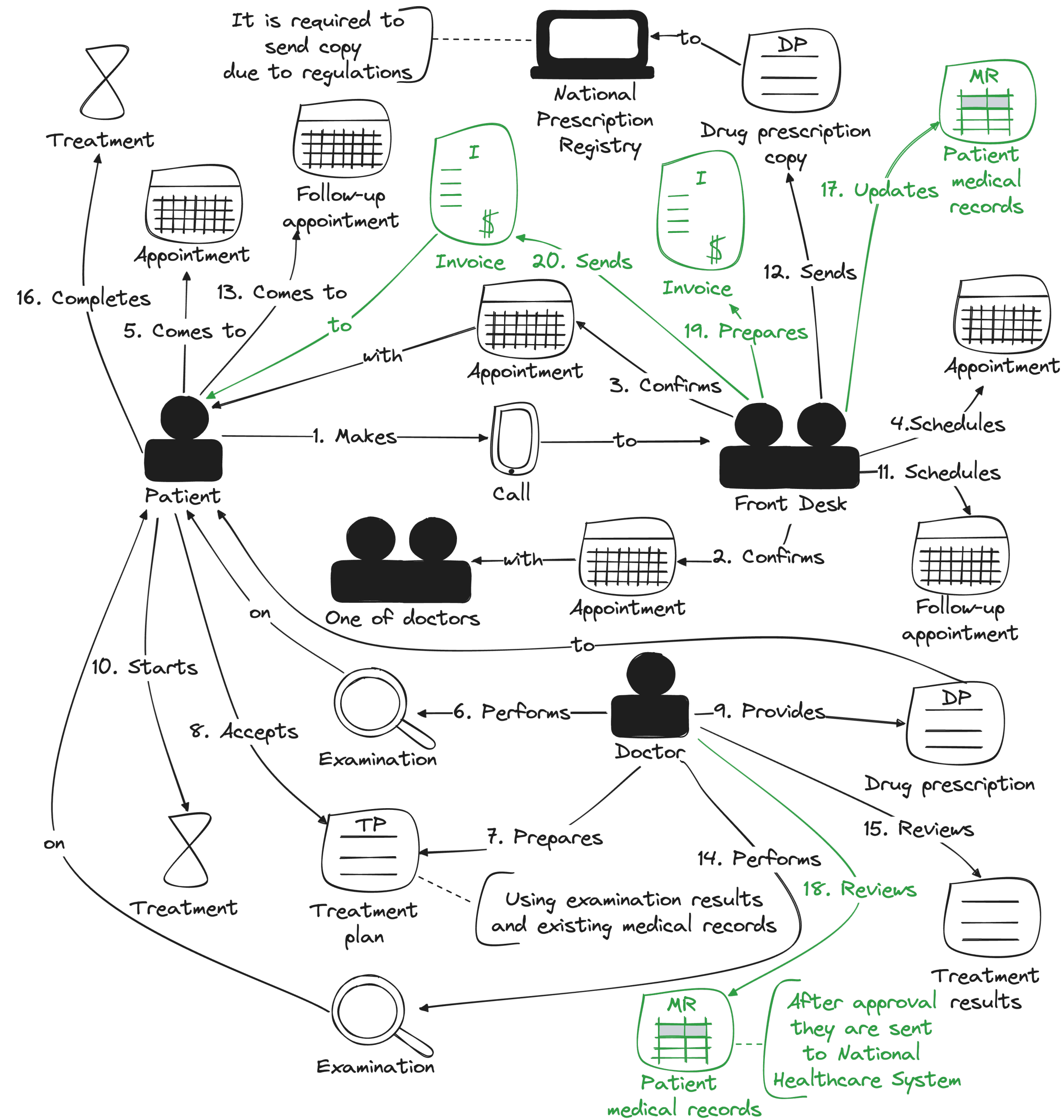
Pass canceled

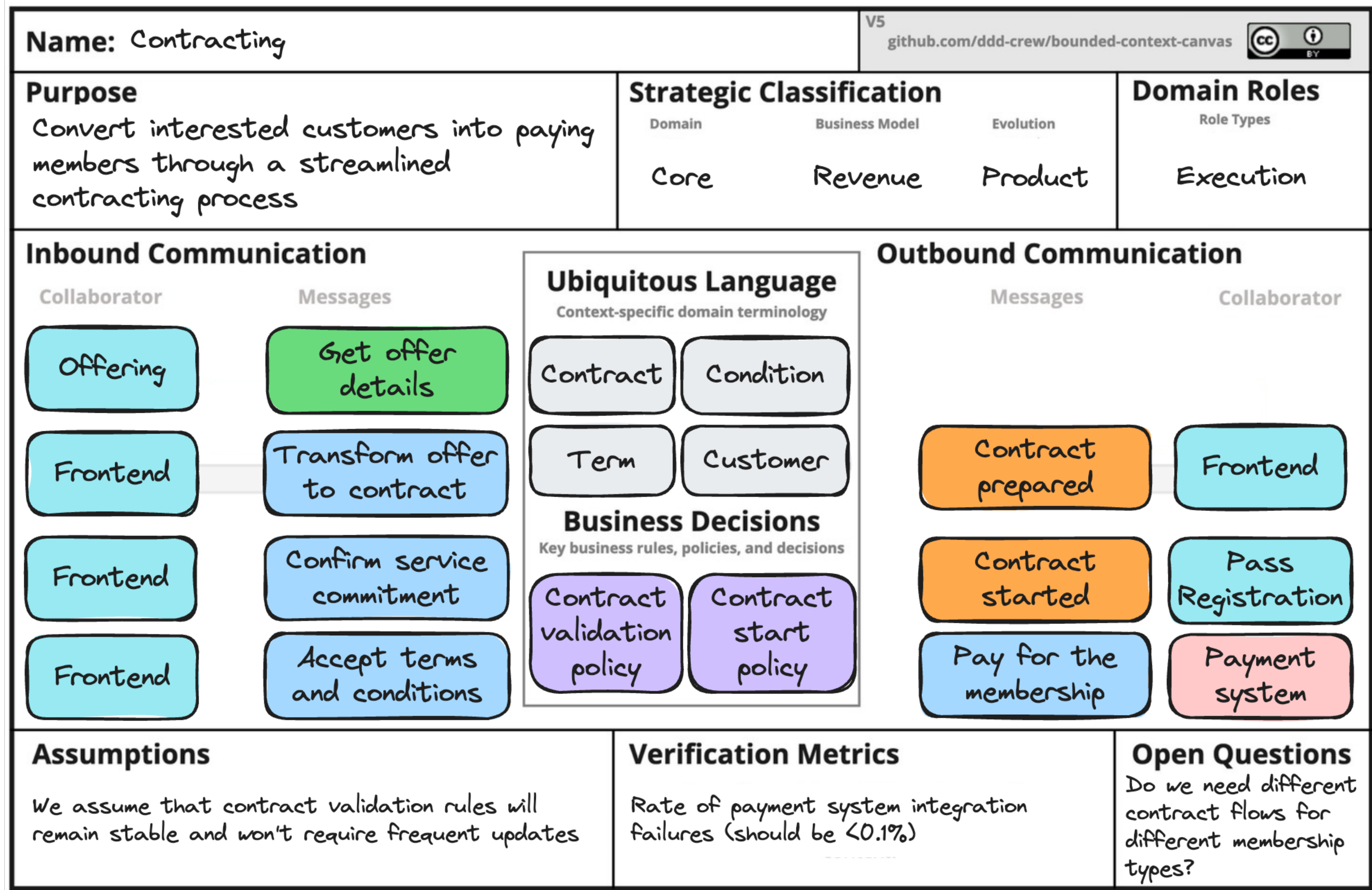
Pass expired

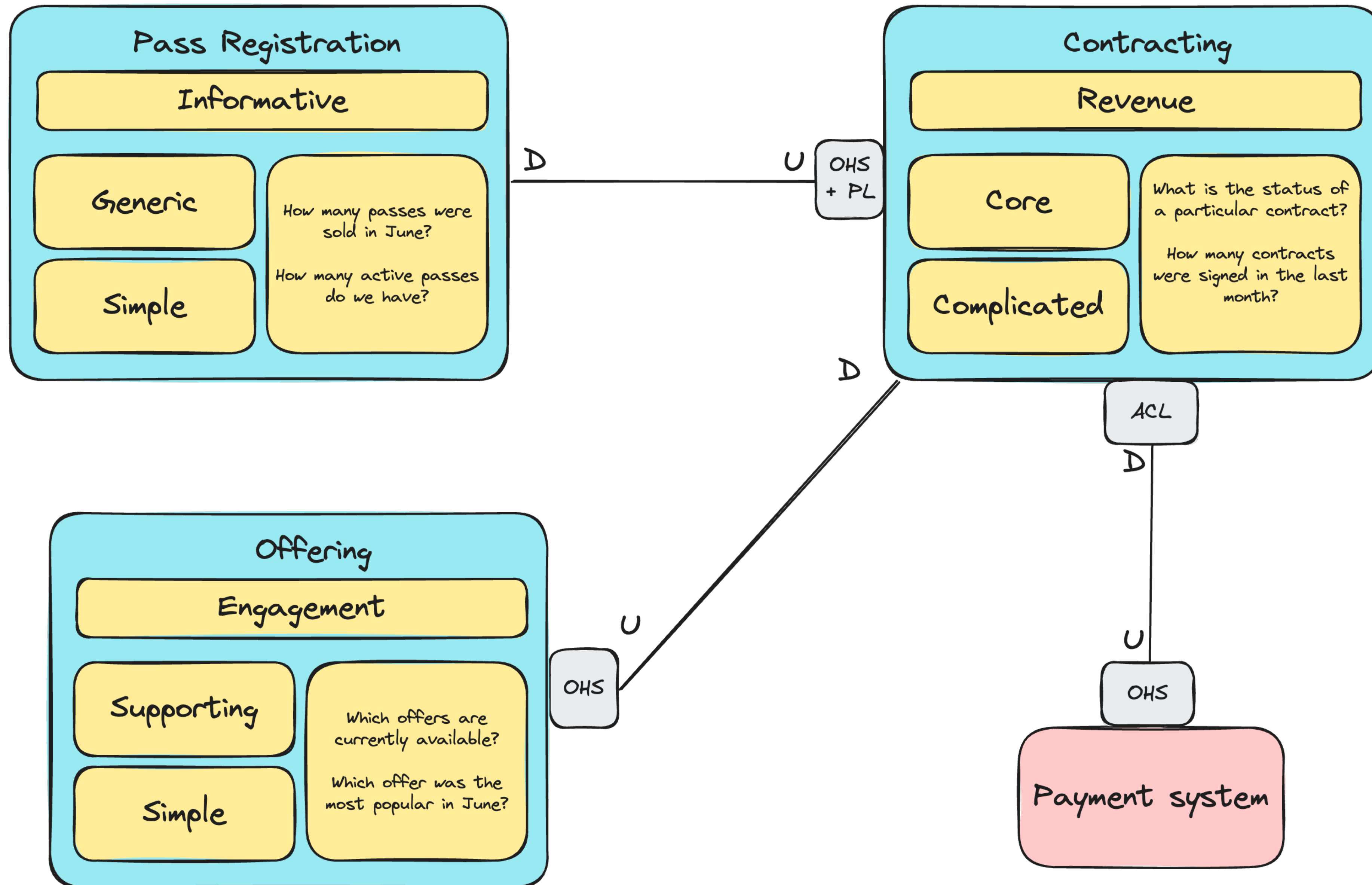
Timeline

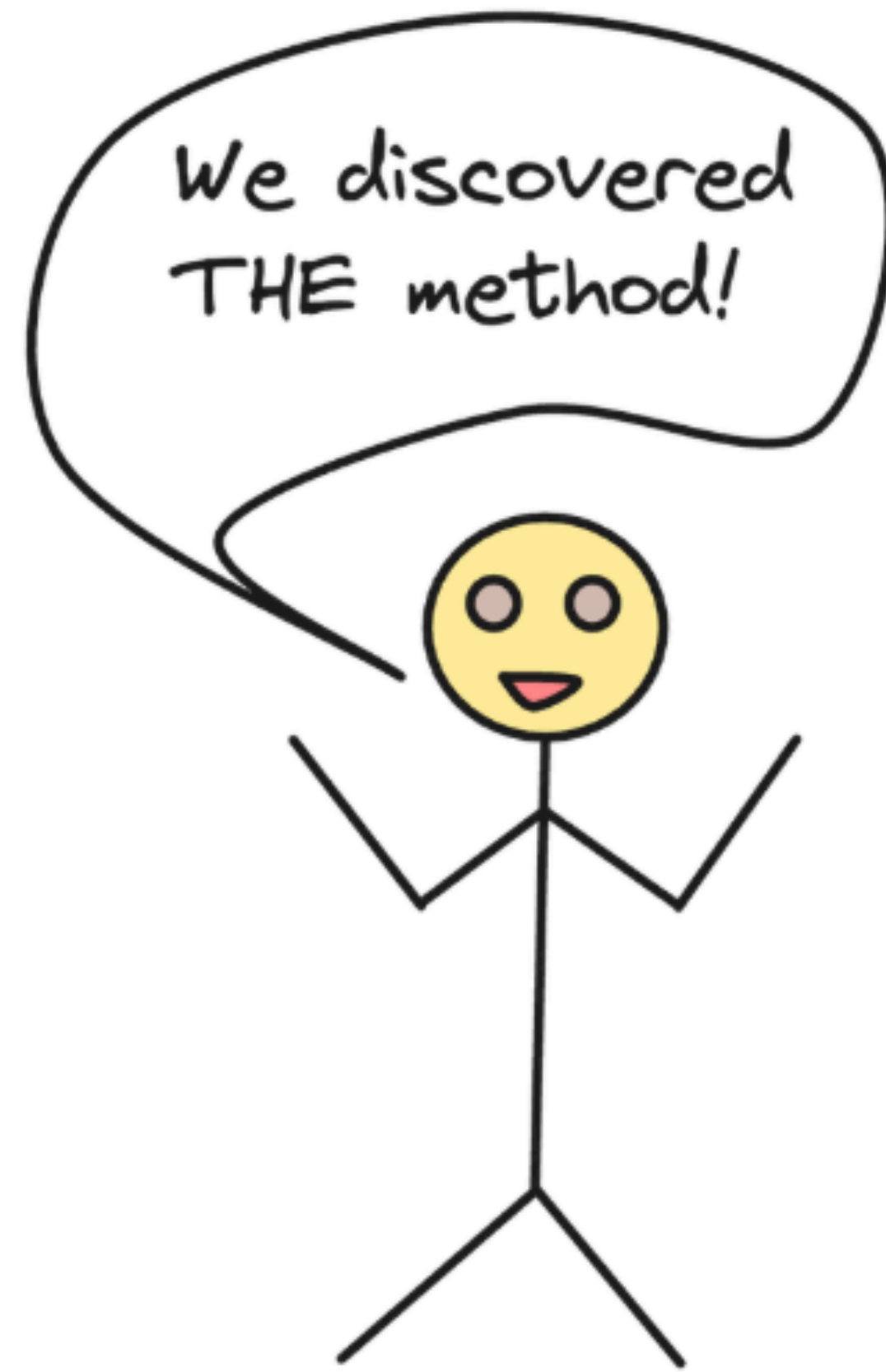










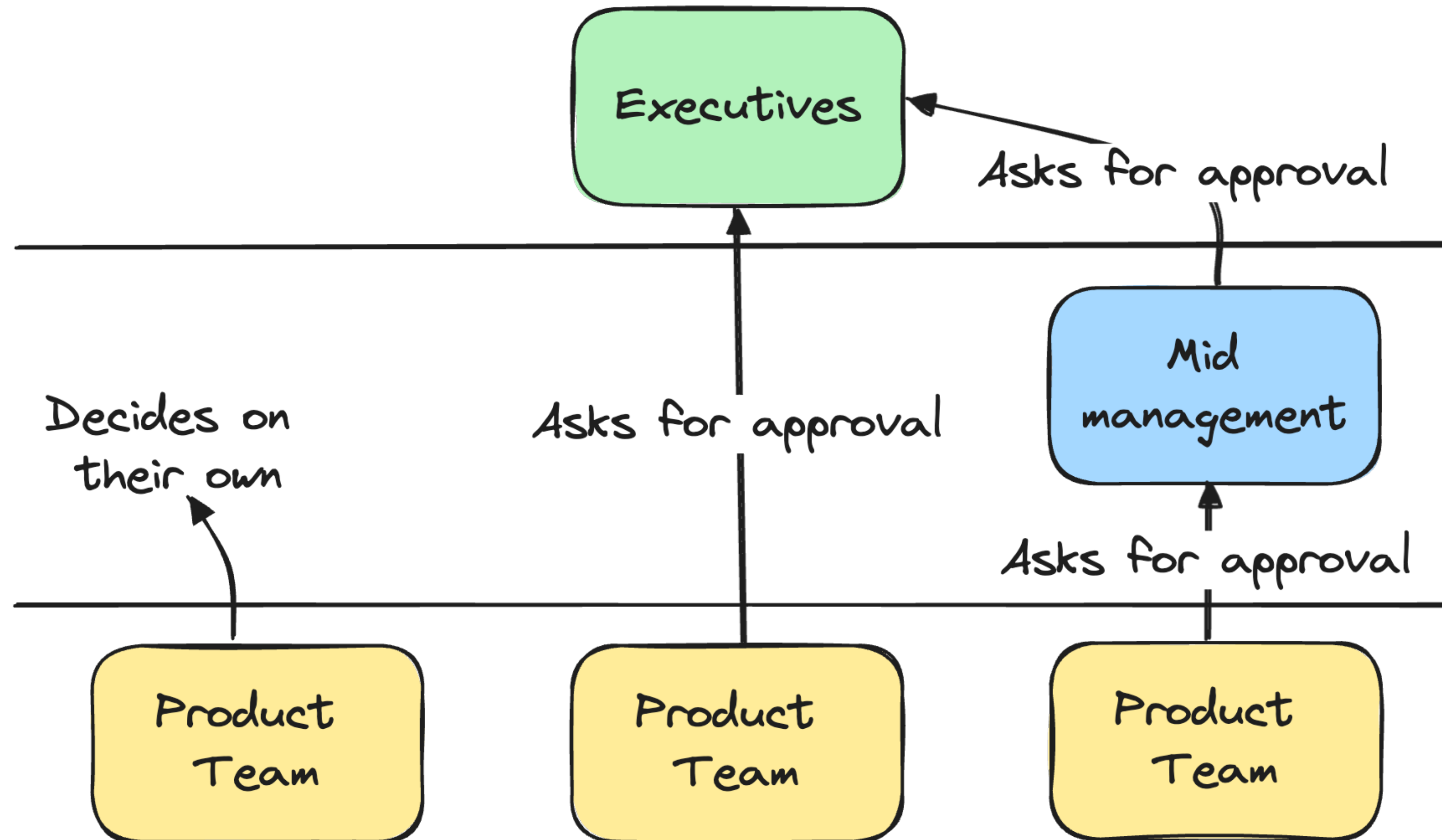


**The best method is the one that fits
YOUR context.**

Not the one that worked for someone else.

Recognize the environment
around you

How are decisions made?



Know your team's reality

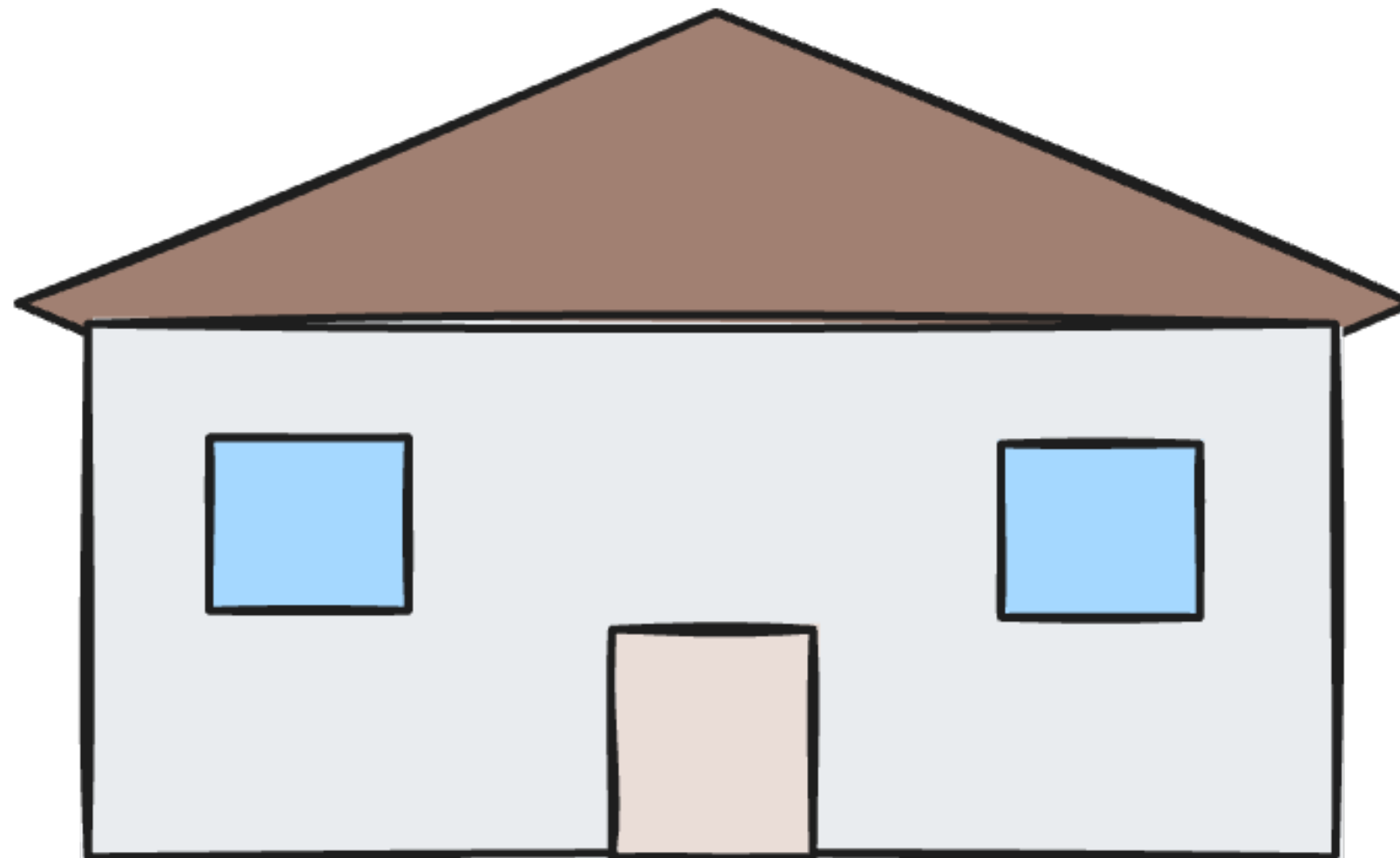
	John	Anna	Susan	Mark	Alex	Total
Java	Expert	OK	Expert	No idea	Expert	4
C#	OK	Expert	No idea	No idea	No idea	2
Javascript	Expert	No idea	Expert	Expert	Expert	4
SQL	Expert	Expert	Expert	OK	Expert	5
NoSQL	No idea	No idea	OK	No idea	Expert	2

Expert
OK
No idea

What are the budget limitations?

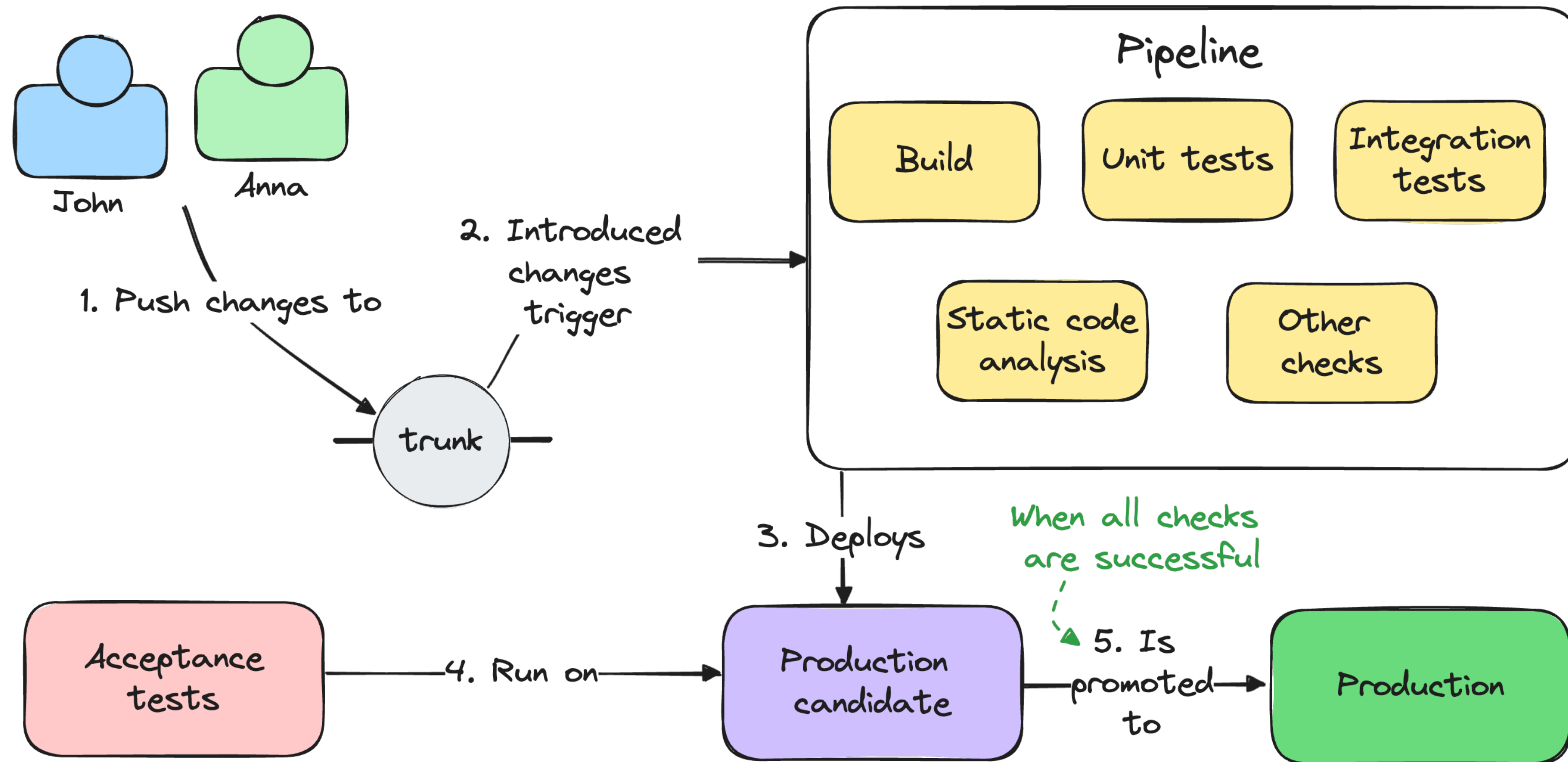
Lesson

Treat it like your
own business

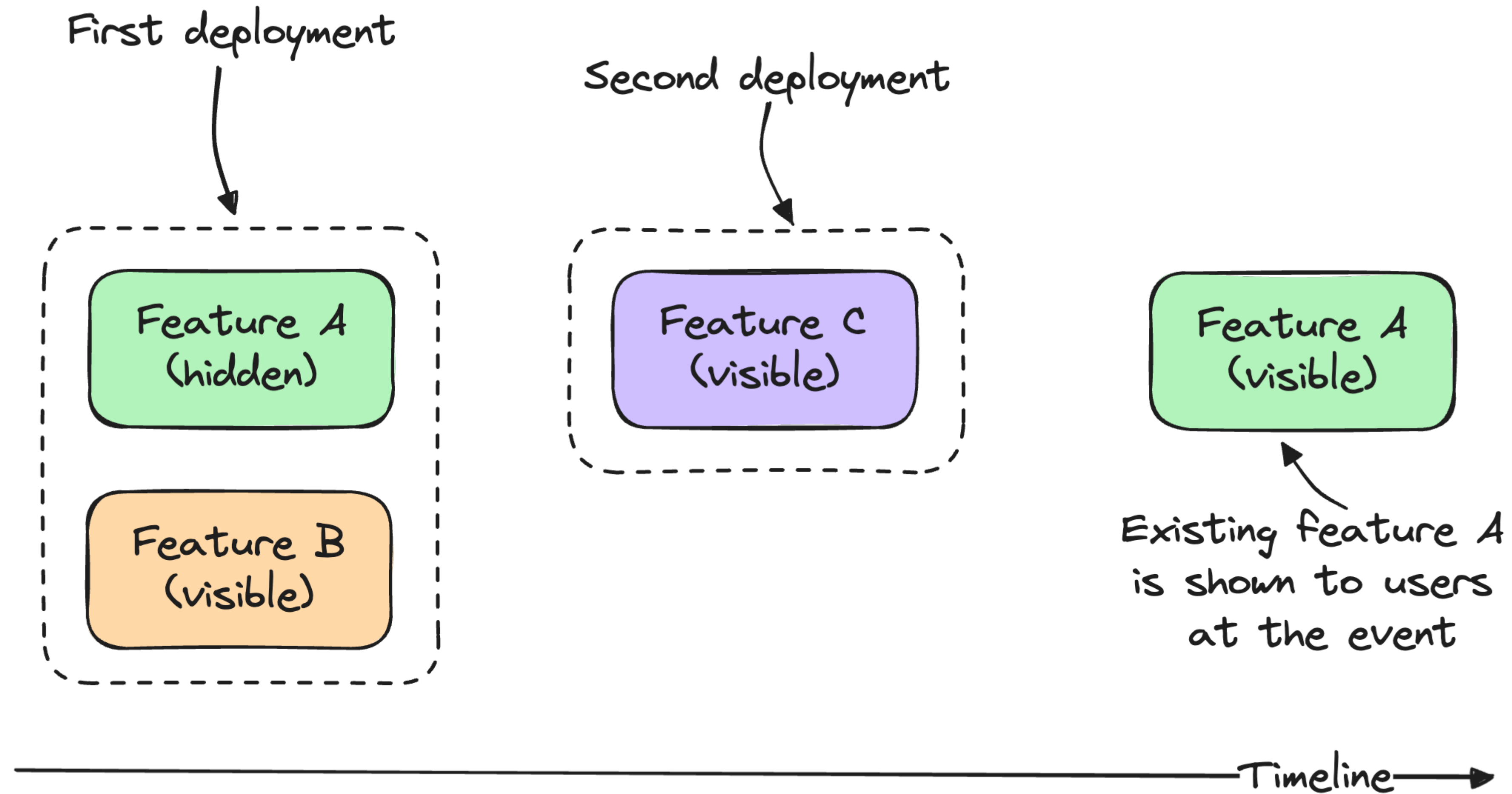


- Are there any other development teams?
- What does existing infrastructure look like?
- What scale and numbers do we expect to serve?
- Do we have a strict deadline?

Deployment strategy

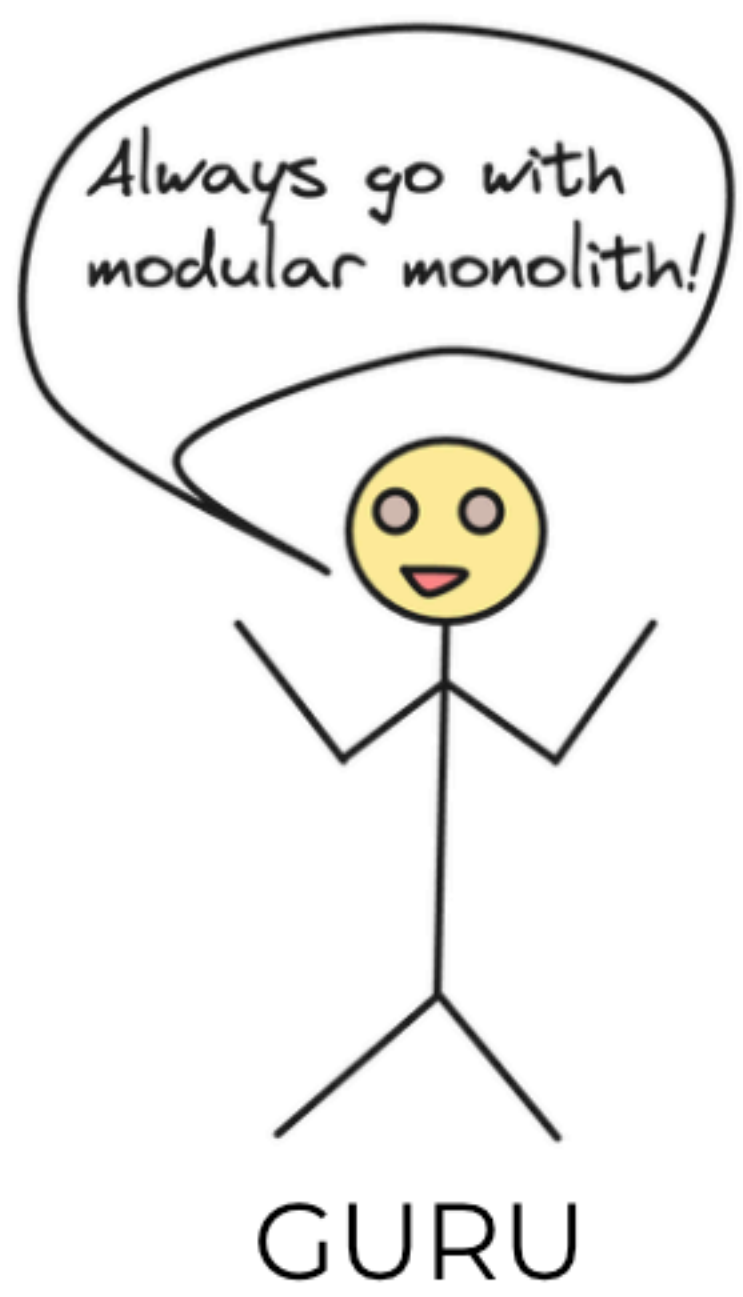
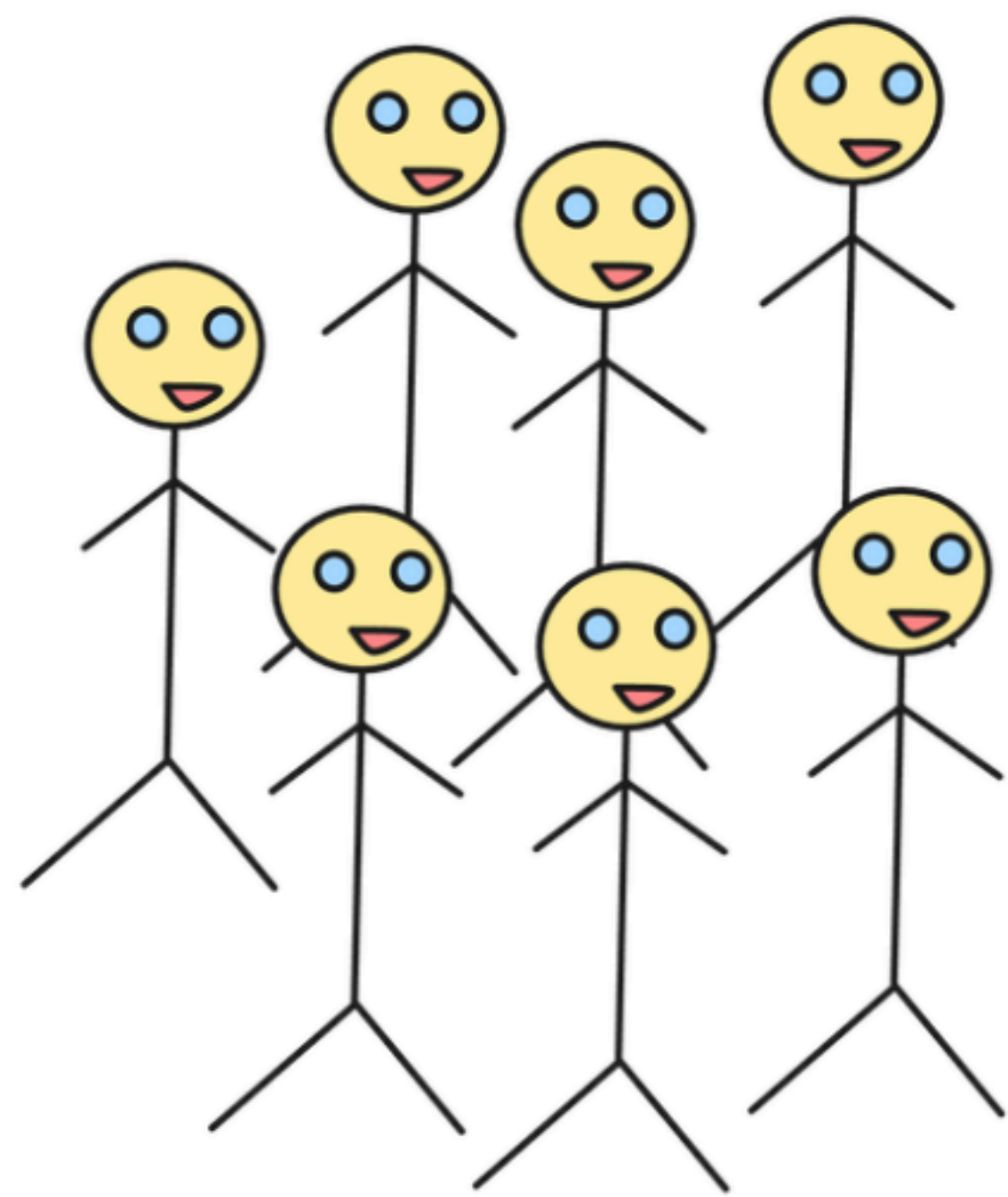


Release strategy



Testing strategy, security

Solution architecture



**“Everything we hear is an opinion, not a fact.
Everything we see is a perspective, not the truth.”**

–Marcus Aurelius

Design your architecture based on
current needs and context.

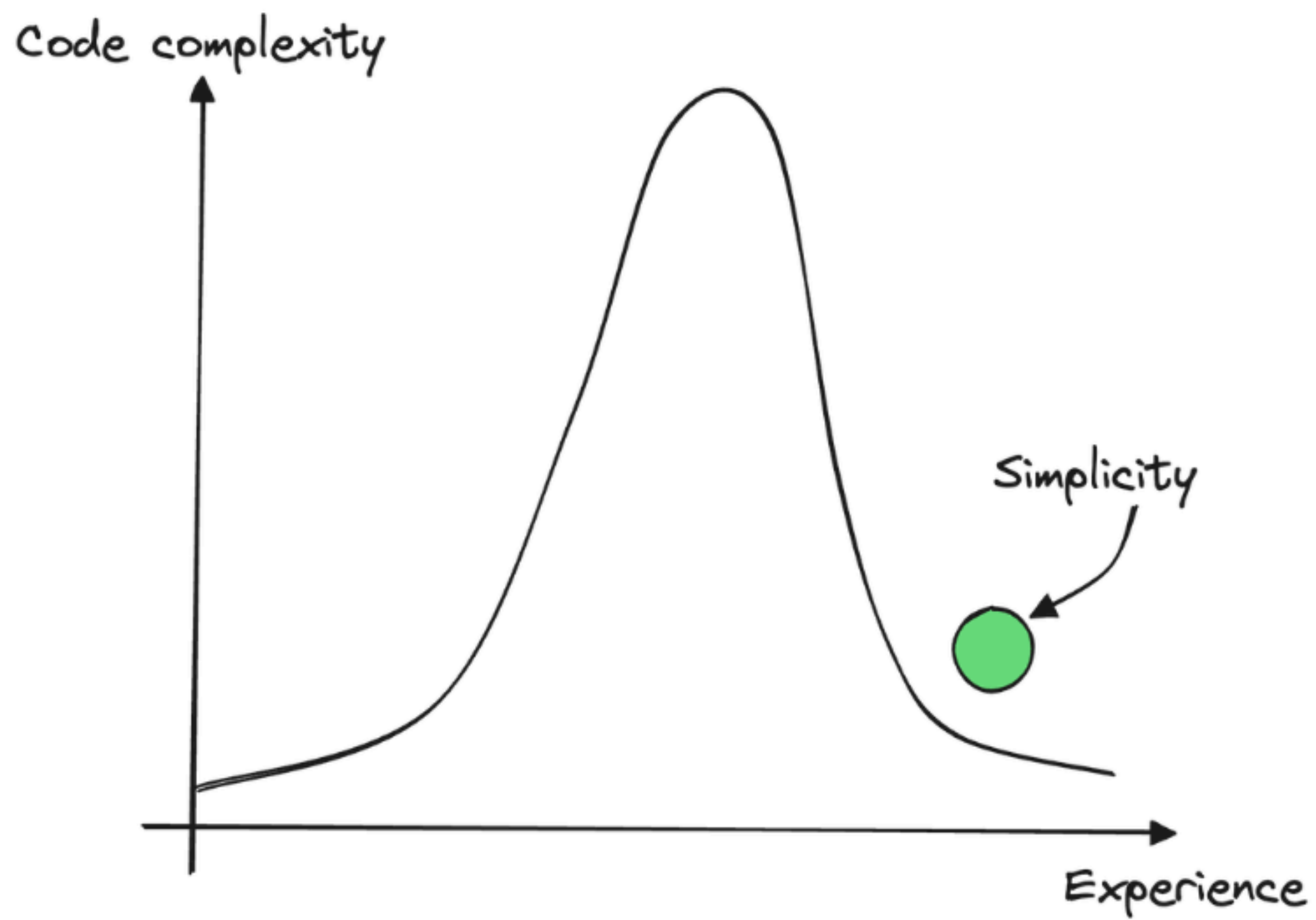
Not a wishful thinking.

Let your system **evolve naturally.**

Four steps of evolution

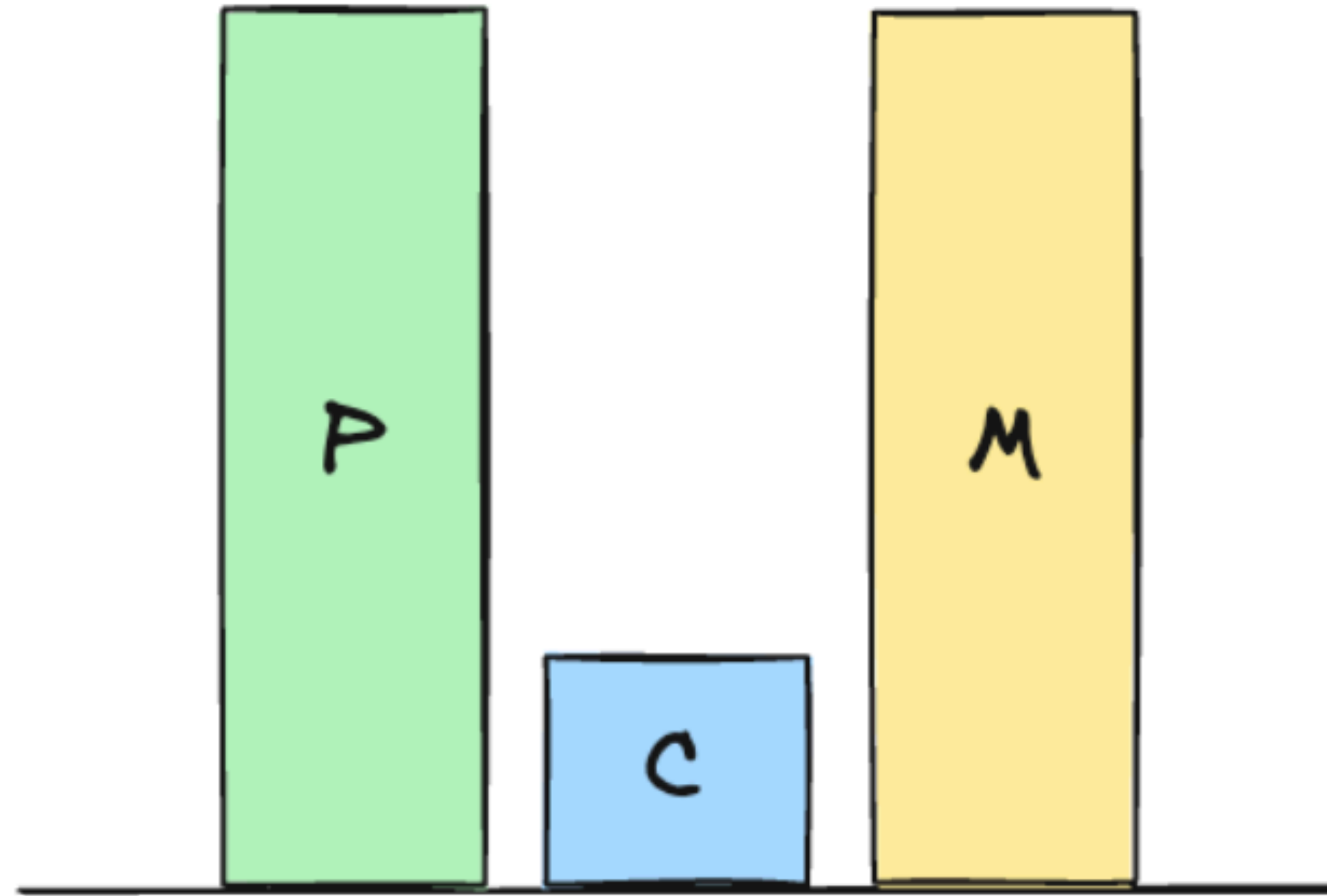
Simplicity

Your main goal is to deliver fast results

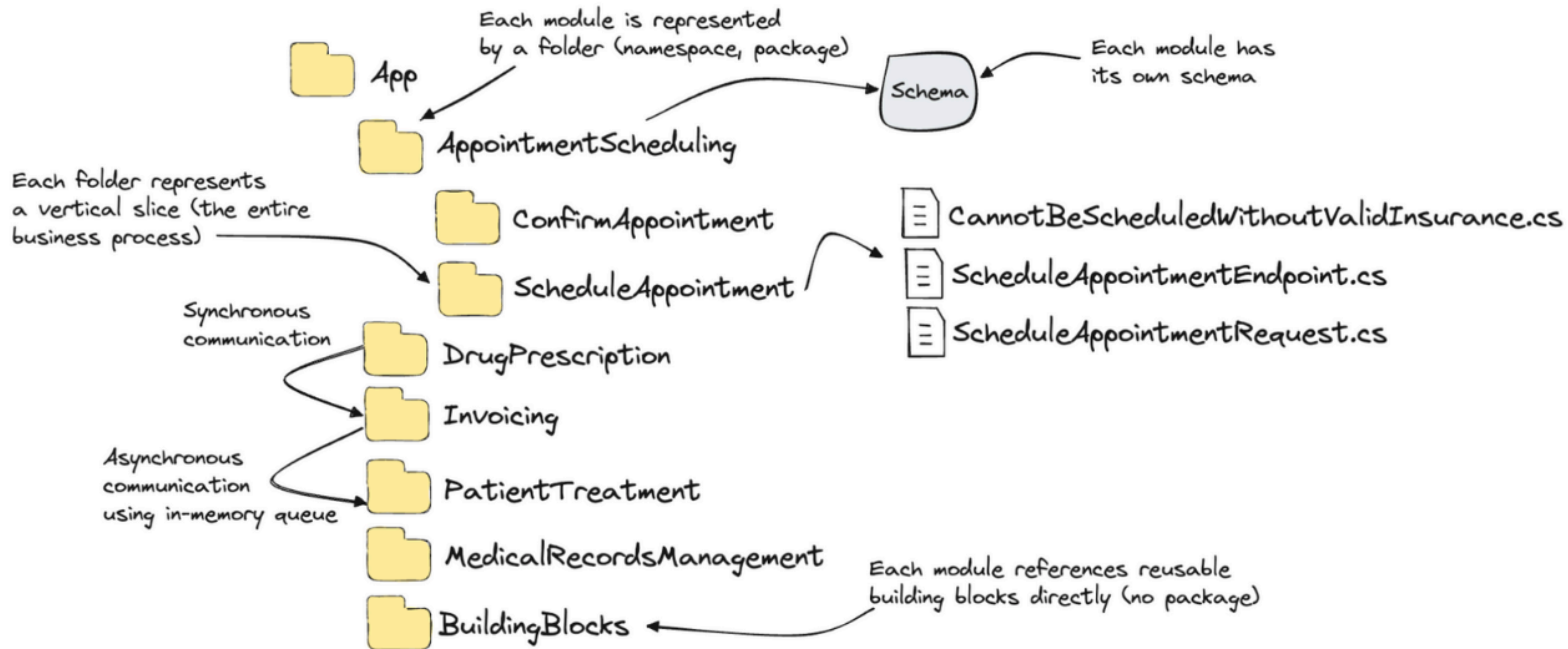


High ↑

Low



- Performance
- Complexity
- Maintainability

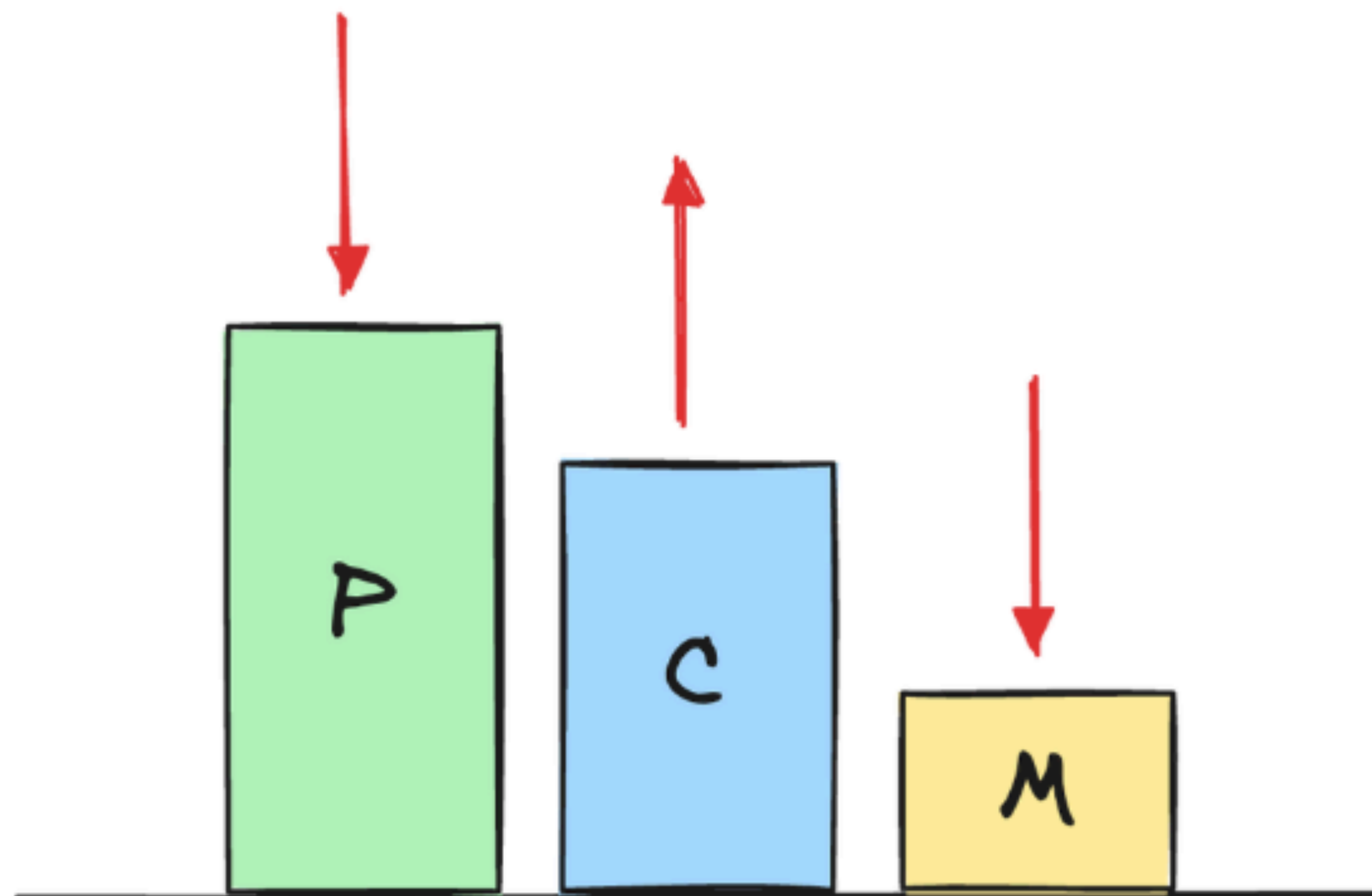


Maintainability

Over time, the complexity of the solution
and the size of the team grows

High ↑

Low



- Performance
- Complexity
- Maintainability



Main



AppointmentScheduling



Invoicing



MedicalRecordsManagement



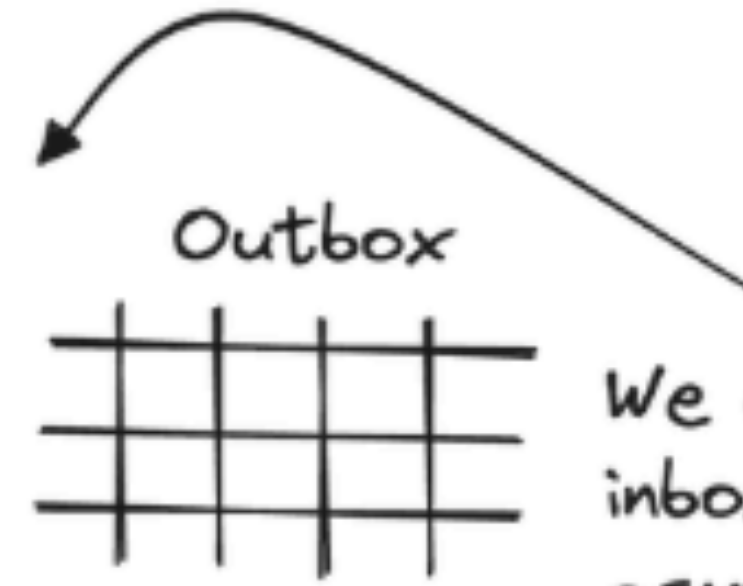
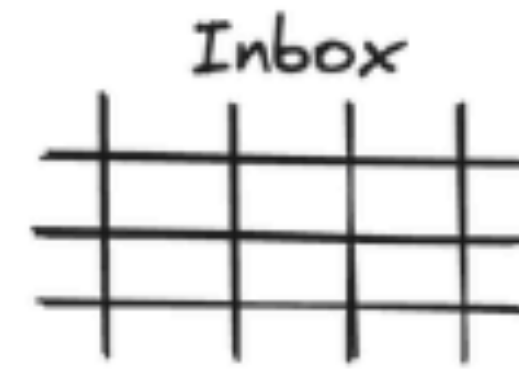
PatientTreatment



DrugPrescription

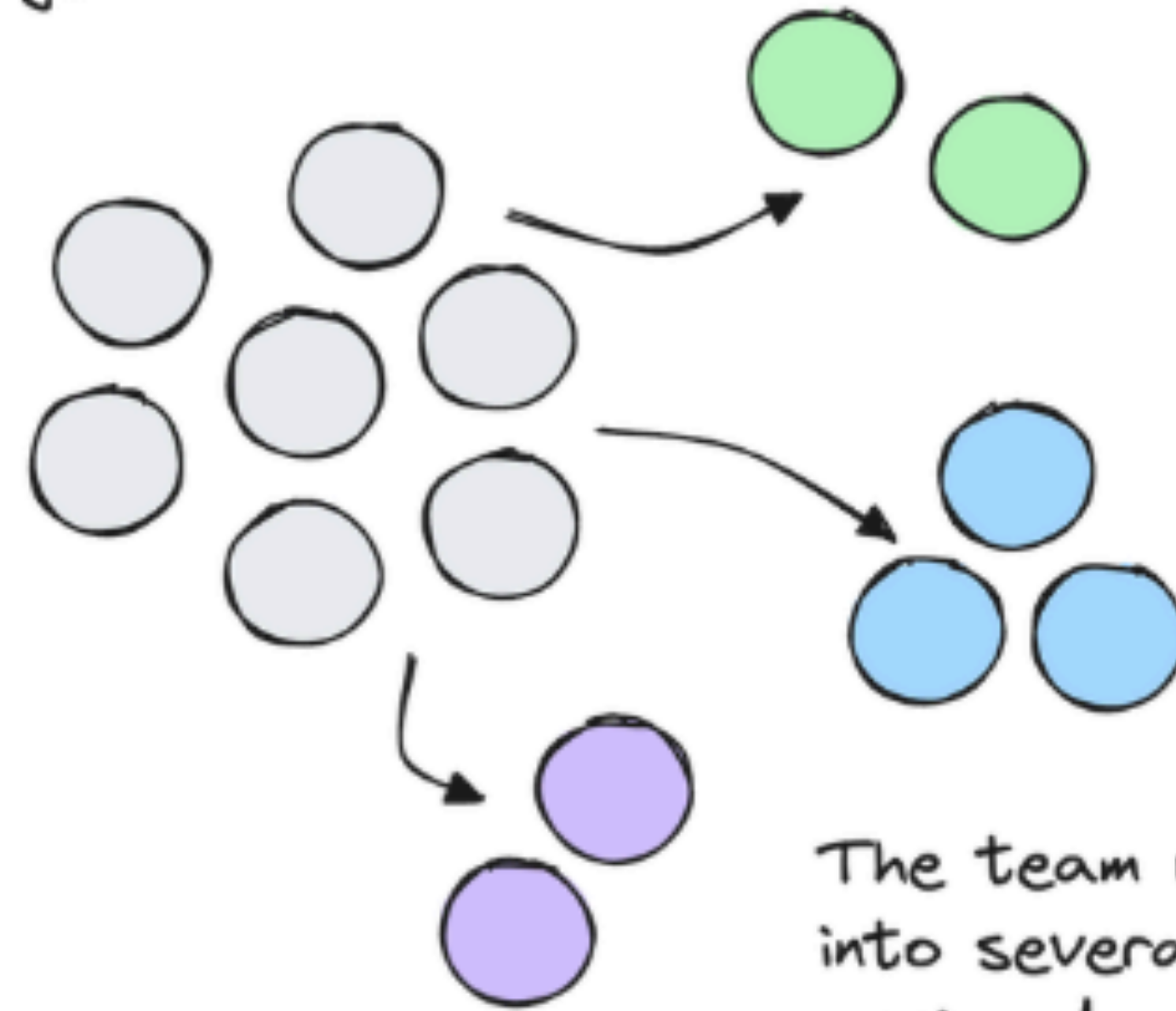


BuildingBlocks



We decided to use
inbox & outbox patterns for
asynchronous communication

Several modules are still
in the main project

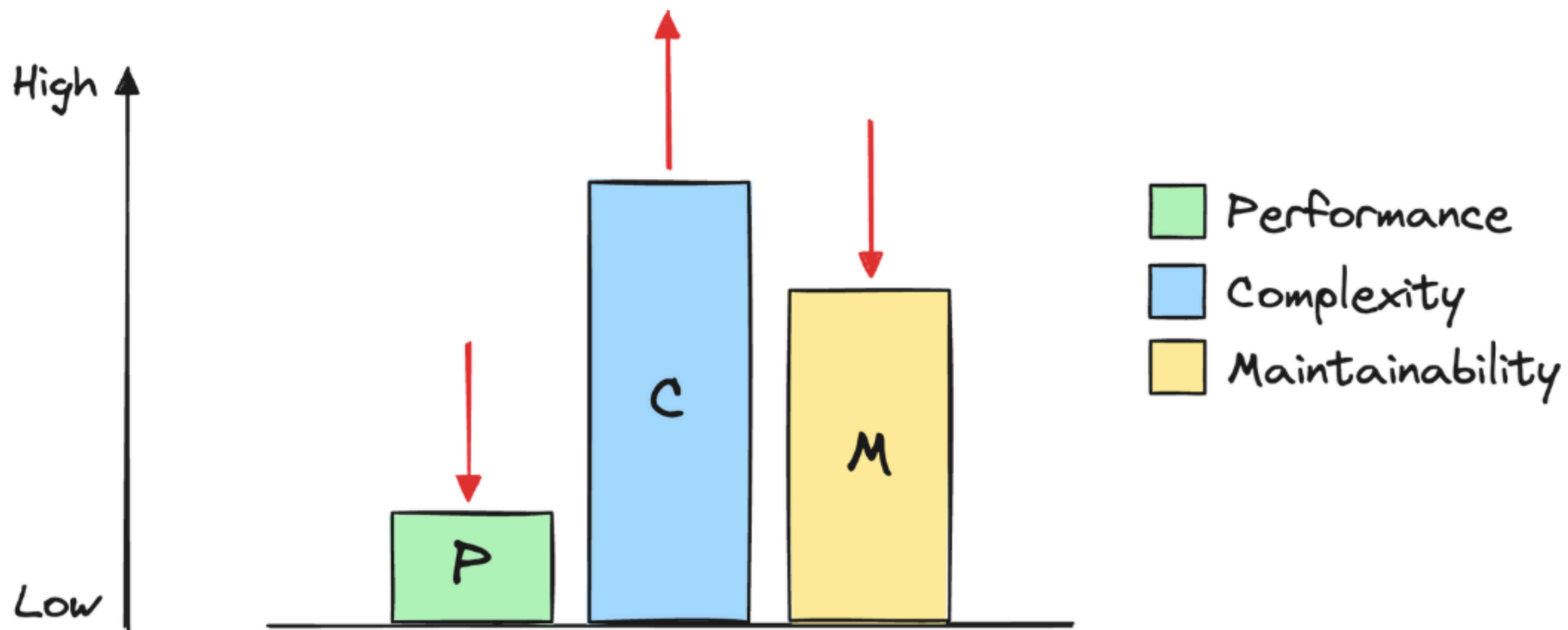


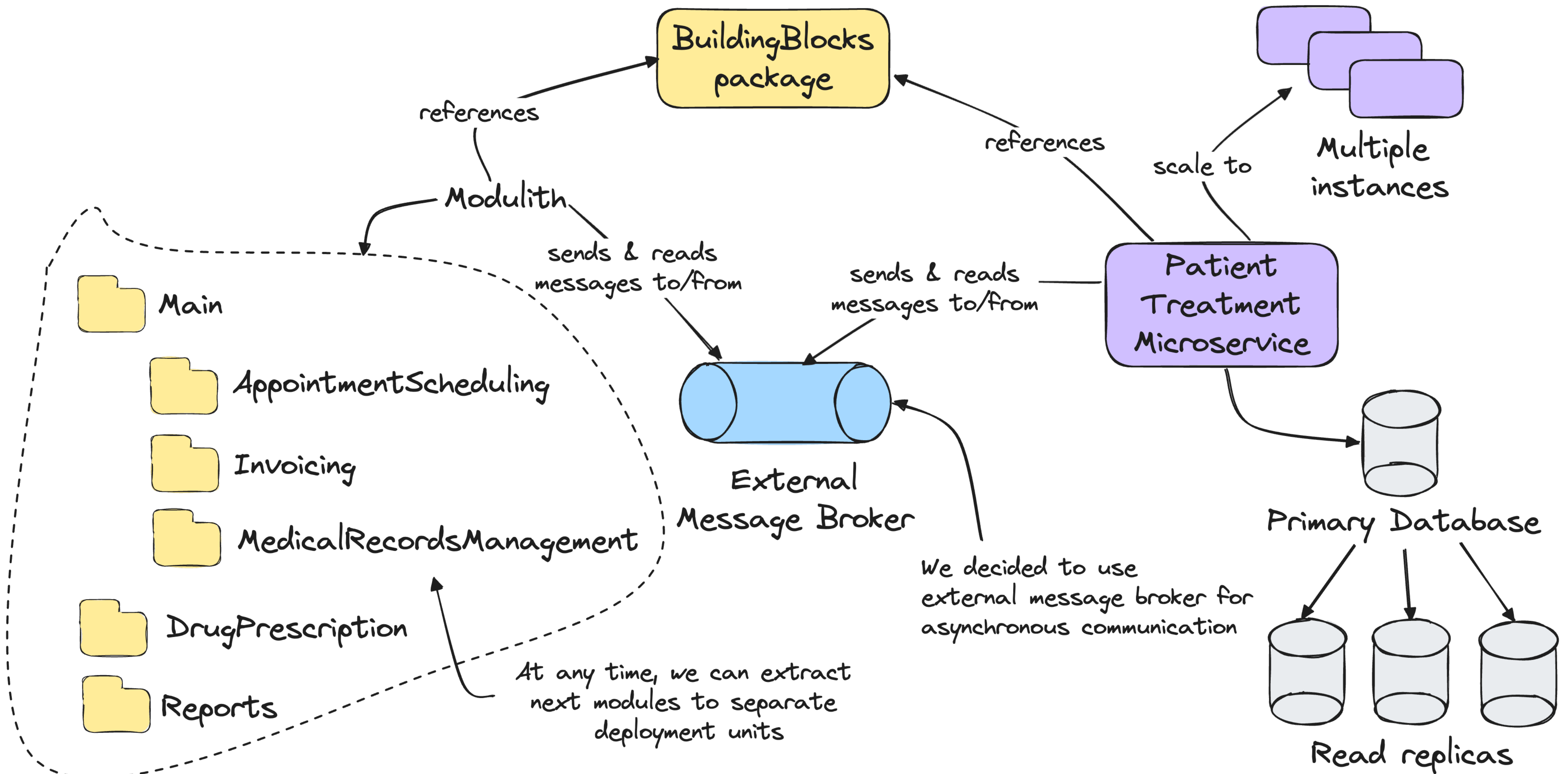
The team was divided
into several others
assigned per module

Two modules are extracted
to separate projects

Growth

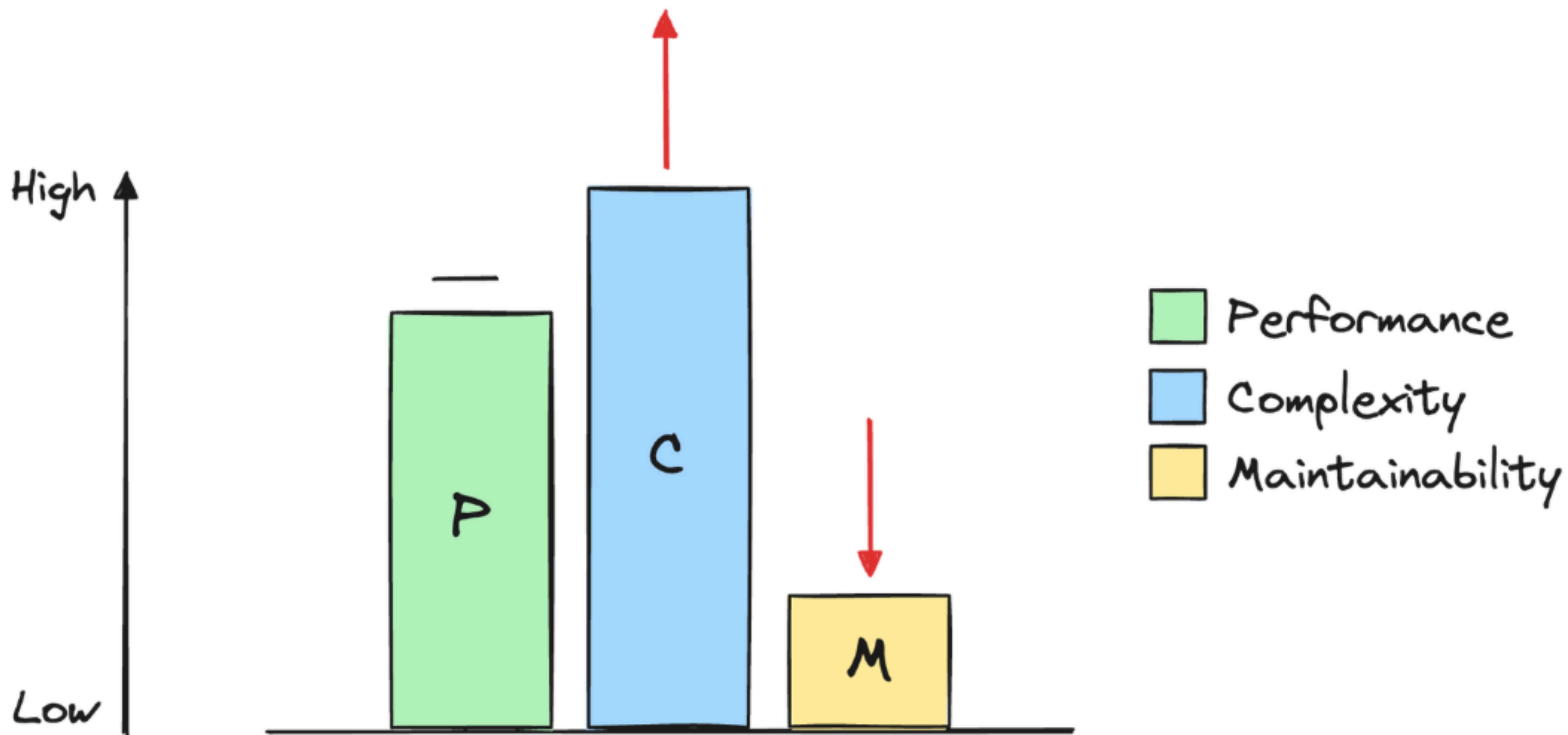
At some point, the growth is so fast that
we can't handle it appropriately.

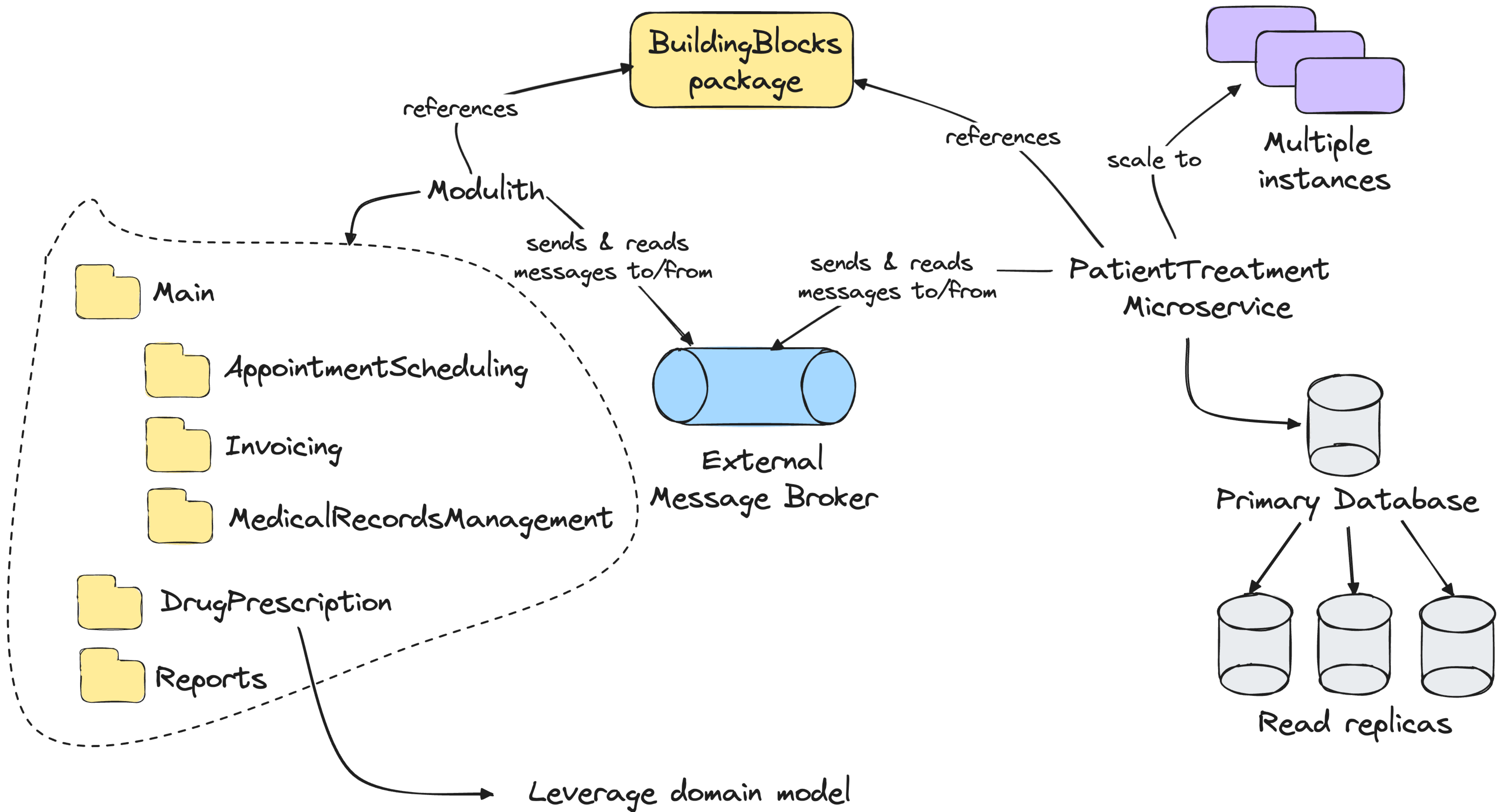




Complexity

When the complexity of the domain increases, revisit your initial choices.





Quick recap

Lessons from the trenches

Cache

Microservices

Incorrect modularization

Logging

The end

